Confidence Calibration under Ambiguous Ground Truth

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Abstract

Confidence calibration assumes a unique ground-truth label per input, an assumption that fails wherever annotators genuinely disagree. Post-hoc calibrators fitted on majority-voted labels can appear well-calibrated against voted labels yet remain substantially miscalibrated against the underlying annotator distribution. This failure is structural: under simplifying assumptions, Temperature Scaling is biased toward temperatures that underestimate annotator uncertainty, with true-label error increasing with annotation entropy. We propose ambiguity-aware post-hoc calibrators that optimize proper scoring rules against the full label distribution, with **Dirichlet-Soft** as the flagship method, substantially improving calibration performance under ambiguous ground truth. For settings where only one-hot voted labels are available, **Label-Smooth Temperature Scaling (LS-TS)** constructs pseudo-soft targets from the model's own confidence without annotator data. Across four benchmarks spanning real multi-annotator distributions (CIFAR-10H, ChaosNLI) and clinically-informed synthetic annotations (ISIC 2019, DermaMNIST), our methods substantially improve calibration performance evaluated by ECE, Brier score, and NLL; Dirichlet-Soft reduces true-label ECE by 59–86% relative to Temperature Scaling and achieves the best or tied-best Brier/NLL in 7 of 8 settings, while LS-TS reduces ECE by 7–78% without any annotator data.

1 Introduction

A well-calibrated classifier $f : \mathcal{X} \to \Delta^K$ produces probability vectors $\hat{p}(x) = f(x)$ such that stated confidence matches empirical accuracy: when $\hat{p}_{\hat{c}}(x) = p$ for the predicted class $\hat{c} = \arg\max_k \hat{p}_k(x)$, the true probability of $\hat{c}$ being correct is exactly p. Formally,

$$\mathbb{P}\big(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p\big) = p, \quad \forall\, p \in [0, 1], \tag{1}$$

where Δ^K denotes the K-dimensional probability simplex. Calibration is critical for safe deployment in medicine [Kompa et al., 2021], autonomous driving [Feng et al., 2022], and other high-stakes settings [Ovadia et al., 2019, Abdar et al., 2021].

The hidden assumption. Equation (1) implicitly assumes a unique ground-truth label Y for every input x. In practice this assumption frequently fails: a skin lesion may be legitimately classified as malignant or benign by different dermatologists; a natural-language premise may or may not entail a hypothesis depending on pragmatic context; low-resolution objects can be categorically ambiguous. In such settings the correct target is not a single label but a *distribution over labels*, $\pi(\cdot \mid x) \in \Delta^K$, representing the underlying ambiguous label distribution induced by genuine annotator disagreement.

The standard practice and its failure. Figure 1 illustrates the core issue: a single input can support multiple plausible labels, and the full label distribution is the object that should be calibrated.

Submitted to 39th Conference on Neural Information Processing Systems (NeurIPS 2025). Do not distribute.

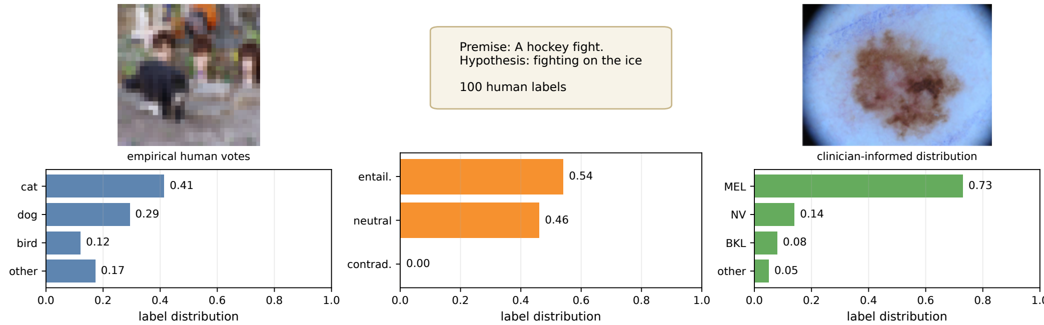


Figure 1: **Left:** a CIFAR-10H image with dispersed human votes (cat/dog/bird), illustrating perceptual ambiguity in low-resolution vision. **Middle:** a ChaosNLI premise–hypothesis pair with split entailment/neutral judgments, illustrating semantic ambiguity. **Right:** an ISIC 2019 melanoma image paired with the clinician-informed label distribution used in our medical experiments, showing clinically plausible MEL/NV confusion. CIFAR-10H and ChaosNLI distributions are empirical human label distributions; the ISIC panel uses the dermatologist confusion model described in Appendix I.

Standard practice nevertheless aggregates annotations by majority vote to obtain a *voted label* y^* and calibrates against it, and in many applications this one-hot voted label is all that is available in the calibration set. We show that this can yield low ECE against voted labels ($\mathrm{ECE}_{\text{voted}}$) while leaving substantial miscalibration against true labels drawn from the ambiguity-aware label distribution $\pi(\cdot \mid x)$ ($\mathrm{ECE}_{\text{true}}$, adopting the voted/true-label terminology of Stutz et al. [2023]). The mechanism is straightforward: ambiguous examples appear underconfident relative to the one-hot voted target, so Temperature Scaling selects a lower T than the ambiguity-aware label distribution requires, amplifying overconfidence. In a controlled experiment (Section 4), all voted-label calibrators fail to correct this mismatch, because the root cause is the voted-label target, not method capacity.

Contributions.

1. **Problem and theory** (Sections 3–5): we formalize true-label calibration under ambiguous labels and show that TS is biased toward overly low temperatures, with true-label miscalibration becoming more severe as annotation entropy increases.
2. **Methods** (Section 6): we propose ambiguity-aware post-hoc calibrators (notably **Dirichlet-Soft**, which achieves the best or near-best calibration performance on most benchmarks) for settings with annotator distributions, and **LS-TS** for the more realistic setting where the calibration set contains only one-hot voted labels.
3. **Evidence** (Sections 4 and 7): a controlled toy example and four real benchmarks show that standard calibrators can worsen true-label calibration, while Dirichlet-Soft reduces $\mathrm{ECE}_{\text{true}}$ by 59–86% and LS-TS by 7–78% consistently across vision and language.

2 Related Work

Confidence calibration. Guo et al. [2017] showed that modern networks are miscalibrated and that TS is an effective post-hoc remedy. Isotonic regression [Zadrozny and Elkan, 2002], Platt scaling [Platt, 1999], and Dirichlet calibration [Kull et al., 2019] extend this to multiclass settings, all assuming a unique ground-truth label. Minderer et al. [2021] showed ViTs are better calibrated than CNNs; Bai et al. [2021] revisit the sufficiency of TS. All share the voted-label assumption; we show this assumption, not method capacity, is the limiting factor. Label smoothing [Szegedy et al., 2016, Müller et al., 2019] applies uniform smoothed targets during training; we instead use annotation-informed label distributions for post-hoc calibration requiring no retraining.

Ambiguous ground truth. Dense annotation datasets [Peterson et al., 2019, Nie et al., 2020] and multi-annotator learning [Uma et al., 2021, Rodrigues and Pereira, 2018] have received growing attention. Baan et al. [2022] show ECE is theoretically ill-defined when humans genuinely disagree and propose instance-level alternatives; our work complements this by providing methods that improve true-label calibration under that setting. We retain $\mathrm{ECE}_{\text{true}}$ as a primary metric for two reasons: (i) it directly quantifies the overconfidence a user experiences when drawing a single label $\tilde{y} \sim \pi(\cdot|x)$ at test time, mirroring realistic deployment; (ii) we complement it with Brier score and NLL, which are strictly proper scoring rules that directly penalise divergence from the full annotator distribution $\hat{\pi}$ and address the distributional concerns raised by Baan et al. [2022]. Gordon et al. [2022] and Plank [2022] highlight epistemological problems with annotation aggregation in NLP. Khurana et al. [2024] study how annotator disagreement can inform calibration in subjective tasks.

Conformal prediction under ambiguous ground truth. Stutz et al. [2023] show that conformal sets calibrated on voted labels undercover the true annotator distribution, the analogous failure for set prediction that our work addresses for point prediction. Calibration under covariate shift [Park et al., 2020, Wang et al., 2020] is orthogonal to our label-ambiguity setting.

3 Problem Formulation

3.1 Setup and Notation

Let $\mathcal{Y} = \{1, \ldots, K\}$ be the label space. A classifier $f : \mathcal{X} \to \Delta^K$ outputs $\hat{p}(x) = \mathrm{softmax}(z(x))$ from pre-softmax logits $z(x) \in \mathbb{R}^K$. The **annotator distribution** $\pi(\cdot \mid x) \in \Delta^K$ denotes the underlying ambiguous label distribution for input x and is estimated from m annotations as $\hat{\pi}_k(x) = \frac{1}{m}\sum_j \mathbf{1}[a_j = k]$. The **voted label** is $y^* = \arg\max_k \hat{\pi}_k(x)$. Standard calibration uses one-hot voted labels. For example, Temperature Scaling [Guo et al., 2017] minimizes

$$\mathcal{L}_{\text{TS}}(T) = -\frac{1}{n}\sum_{i=1}^{n} \log \mathrm{softmax}(z_i/T)_{y_i^*}. \tag{2}$$

3.2 True-Label Calibration

Following Stutz et al. [2023], we distinguish the voted label y^* from the **true label** $\tilde{y} \sim \pi(\cdot \mid x)$, drawn from the underlying ambiguous label distribution.

Definition 1 (True-Label Calibration)**.** *f is **true-label calibrated** if for all $p \in [0,1]$: $\mathbb{P}(\tilde{Y} = \hat{c}(X) \mid \hat{p}_{\hat{c}}(X) = p) = p$, where $\hat{c}(x) = \arg\max_k \hat{p}_k(x)$ and $\tilde{Y} \sim \pi(\cdot \mid X)$.*

The **voted-label ECE** $\mathrm{ECE}_{\text{voted}}$ is the standard ECE against y^*. The **true-label ECE** $\mathrm{ECE}_{\text{true}}$ averages ECE over 100 draws $\tilde{y}_i \sim \pi(\cdot \mid x_i)$ ($B = 15$ equal-width bins). Throughout the paper we compare these two views directly, since ambiguous examples can look well-calibrated under voted labels while remaining overconfident under the underlying ambiguous label distribution.

4 Motivating Example

Setup. We construct a 3-class 2D Gaussian dataset with three clusters: $x \sim \mathcal{N}((-3.2, 1.1), \mathrm{diag}(0.60, 0.45))$ for class 0 with $\pi = [1, 0, 0]$; $x \sim \mathcal{N}((0, 0), \mathrm{diag}(1.15, 0.75))$ for the ambiguous cluster with $\pi = [0, 0.70, 0.30]$; and $x \sim \mathcal{N}((3.2, -1.1), \mathrm{diag}(0.60, 0.45))$ for class 2 with $\pi = [0, 0, 1]$. The voted label for the middle cluster is always class 1 because $0.70 > 0.30$. A 2-layer MLP (64 hidden units per layer) is trained on voted labels, and all calibrators are fitted on a held-out calibration split using the same voted labels.

Results. Figure 2 shows that this mismatch is a structural consequence of the voted-label target. TS reduces $\mathrm{ECE}_{\text{voted}}$ from 3.25% to 1.11%, but *increases* $\mathrm{ECE}_{\text{true}}$ from 6.26% to 9.64%. Platt scaling behaves similarly ($\mathrm{ECE}_{\text{voted}} = 0.68\%$, $\mathrm{ECE}_{\text{true}} = 9.70\%$), and Histogram Binning likewise ($\mathrm{ECE}_{\text{voted}} = 1.32\%$, $\mathrm{ECE}_{\text{true}} = 9.65\%$). Figure 2**(f)** further shows that this residual error is concentrated in the ambiguous cluster for all three methods. TS sets $T = 0.62 < 1$, boosting confidence toward the voted target and thereby pushing the model *further* from the true ambiguous label distribution. The failure is therefore not specific to TS: standard post-hoc methods are useful for $\mathrm{ECE}_{\text{voted}}$, but the voted-label target makes them ineffective for $\mathrm{ECE}_{\text{true}}$.

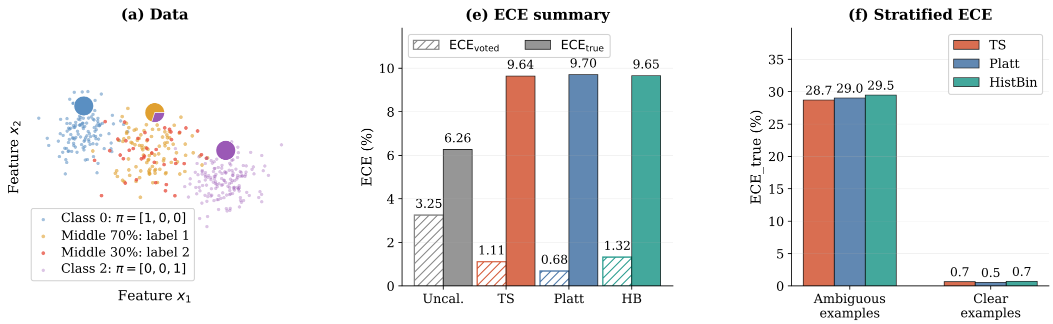


Figure 2: **Motivating example: all standard calibration methods fail. (a)** Toy dataset generation: only the middle Gaussian cluster is ambiguous, with $\pi = [0, 0.70, 0.30]$; orange points are drawn as label 1 (70%) and red points as label 2 (30%), while all receive the same voted label 1. **(e)** Summary of the toy results: all three voted-label calibrators (TS, Platt, Histogram Binning) lower $\mathrm{ECE}_{\mathrm{voted}}$ but *increase* $\mathrm{ECE}_{\mathrm{true}}$, so voted-label evaluation masks the failure. **(f)** Stratified $\mathrm{ECE}_{\mathrm{true}}$ for TS, Platt, and Histogram Binning: ambiguous examples are those from the middle Gaussian cluster (where annotators disagree); clear examples are those from the two unambiguous clusters (class 0 and class 2). The residual error is concentrated in ambiguous examples for all three methods.

5 Theory: Why Standard Calibration Fails

We now formalize, under simplifying assumptions, two complementary aspects of the failure of voted-label calibration under ambiguity: the *direction* of the TS bias (Proposition 1) and its *scaling* with label ambiguity (Proposition 2). Both propositions characterize *pointwise* miscalibration on individual examples; they describe when and why a voted-label calibrator misfits individual ambiguous inputs, rather than providing a theorem about population-level ECE. Empirical validation of Proposition 2 is provided in Appendix J.

Proposition 1 (Direction of TS Bias)**.** *Consider a calibration set containing an ambiguous cluster in which the voted label y^* equals the majority class for every example, but the true annotator probability of y^* is $\pi_{y^*}(x) = q < 1$. Under the assumption that the model is already reasonably accurate on this cluster ($\mathrm{softmax}(z_i)_{y^*} > q$ for all i in the cluster), the voted-label loss exerts a downward pull on T for these examples, whereas a soft-label loss with target q exerts an upward pull. Consequently, $T^*_{TS} < T^*_{soft}$: TS selects a lower temperature than ambiguity-aware calibration requires. In a setting where the model is already overconfident ($\mathrm{softmax}(z_i)_{y^*} \gg q$), this manifests as $T^*_{TS} < 1$; more generally, both optima may exceed 1, but the gap $T^*_{soft} - T^*_{TS} > 0$ persists, consistent with our experimental findings (e.g., $T_{TS} = 2.03$ vs. $T_{SLTS} = 3.18$ on CIFAR-10H ResNet-50).*

Proof sketch. On the ambiguous cluster, all voted labels are y^*, so the voted-label loss (2) is minimized by pushing $\mathrm{softmax}(z_i/T)_{y^*} \to 1$, i.e. the loss decreases as T decreases. Hence the ambiguous examples exert a downward gradient on T. A soft-label loss with target $q < 1$ is instead minimized at $\mathrm{softmax}(z_i/T^*)_{y^*} \approx q$, requiring a higher T^* whenever $\mathrm{softmax}(z_i)_{y^*} > q$. The global TS optimum balances this downward pull against the upward pull from unambiguous examples, but remains lower than the ambiguity-aware optimum. Full proof in Appendix B. $\square$

Proposition 2 (True-Label Miscalibration and Annotation Entropy)**.** *Let $H(x) = -\sum_k \pi_k(x) \log \pi_k(x)$ denote the annotation entropy of x. Assume* (i) *the model's predicted confidence $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$ (similar accuracy across ambiguity levels), and* (ii) *$\pi_{y^*}(x)$ is a decreasing function of $H(x)$ (greater entropy implies lower majority-class probability). Under assumptions* (i) *and* (ii)*, the expected per-example true-label calibration error of voted-label-calibrated predictions is non-decreasing in $H(x)$.*

Proof sketch. For unambiguous examples ($H(x) = 0$), $\pi(\cdot \mid x)$ is one-hot, so true-label and voted-label evaluation coincide and the per-example error is $|\hat{p}_{\hat{c}}(x) - 1|$. As $H(x)$ increases, assumption (ii) gives $\pi_{y^*}(x) < 1$, introducing a gap between the voted-label target and the true annotator probability. Because TS is optimised against one-hot voted labels, it targets $\hat{p}_{y^*}(x) \approx 1$ for all examples regardless of their annotation entropy; hence $\hat{p}_{\hat{c}}(x) > \pi_{y^*}(x)$ for ambiguous examples. The true-label calibration error is then $\hat{p}_{\hat{c}}(x) - \pi_{y^*}(x)$, which by assumption (i) (fixed $\hat{p}_{\hat{c}}$) increases as $\pi_{y^*}(x)$ falls with $H(x)$. $\square$

6 Methods

All methods in this paper are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require only a calibration set equipped with annotator distributions or individual annotations. No retraining or architectural modification is needed. Training-time approaches such as soft-label training [Szegedy et al., 2016, Müller et al., 2019], mixup [Thulasidasan et al., 2019], and multi-annotator learning [Uma et al., 2021, Rodrigues and Pereira, 2018] can also improve calibration under ambiguity, but they require access to the training pipeline and are orthogonal to post-hoc calibration; we do not compare against them here.

6.1 Ambiguity-Aware Calibration Methods

Soft-Label Temperature Scaling (SLTS). SLTS minimizes the cross-entropy between the calibrated output and the empirical label distribution:

$$\begin{aligned}\mathcal{L}_{\text{SLTS}}(T) &= -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}\hat{\pi}_k(x_i)\,\log\operatorname{softmax}(z_i/T)_k \\ &= \frac{1}{n}\sum_{i=1}^{n}\operatorname{KL}\big(\hat{\pi}(x_i)\,\|\,\operatorname{softmax}(z_i/T)\big) + \text{const.}\end{aligned} \tag{3}$$

SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator distribution.

Proposition 3 (Correctness of distributional target)**.** *The per-example loss $\ell(q;\hat{\pi}) = -\sum_k \hat{\pi}_k \log q_k$ is a strictly proper scoring rule over $q \in \Delta^K$: it is uniquely minimised at $q = \hat{\pi}(x)$ for each x. Consequently, within the constrained family $\{\operatorname{softmax}(z/T) : T > 0\}$, $\mathcal{L}_{SLTS}$ identifies the temperature T^* that minimises $\operatorname{KL}(\hat{\pi}\|\operatorname{softmax}(z/T^*))$, targeting the correct distributional objective, unlike $\mathcal{L}_{TS}$, which targets the one-hot voted label. Note that T^* need not achieve $\operatorname{softmax}(z/T^*) = \hat{\pi}$ exactly within this one-parameter family; the proper scoring rule property guarantees correctness of the* target*, not achievability within the constrained family.*

Vector Scaling (VS). Extends SLTS with per-class temperatures $\mathbf{T} = (T_1, \ldots, T_K)$, accommodating datasets where some classes are systematically more ambiguous than others.

Distributional Isotonic Regression (IR-Soft). Fits a monotone mapping from predicted confidence to mean annotator accuracy via the Pool Adjacent Violators Algorithm (PAVA), using the empirical label distribution as target.

SoftPlatt. The diagonal affine transform $W \cdot z + b$ (same structure as Platt) fitted against the annotator distribution via KL divergence. This isolates the contribution of the distributional target from any architectural difference.

Dirichlet-Soft. The full $K \times K$ affine $Wz + b$ fitted by minimizing $\operatorname{KL}(\hat{\pi}(x)\|\operatorname{softmax}(Wz + b))$ with ODIR regularization ($\lambda = 10^{-3}$) [Kull et al., 2019]. This is the most expressive parametric ambiguity-aware calibrator.

6.2 Annotation-Free Calibration

The methods above require annotator distributions $\hat{\pi}(x_i)$ at calibration time. When only voted labels y_i^* are available, we propose **Label-Smooth Temperature Scaling (LS-TS)**, which constructs a pseudo-target distribution from the model's own predictions without any additional annotations.

Let $\bar{\varepsilon} = \frac{1}{n}\sum_{i=1}^{n}\big(1 - \hat{p}_{y_i^*}(x_i)\big)$ be the mean complement of the model's voted-class confidence on the calibration set. Define the pseudo-target distribution

$$\tilde{\pi}_i^{\text{LS}} = (1 - \bar{\varepsilon})\, e_{y_i^*} + \tfrac{\bar{\varepsilon}}{K}\,\mathbf{1}, \tag{4}$$

and minimize $\mathcal{L}_{\text{SLTS}}$ (Eq. 3) with these targets. The global smoothing weight $\bar{\varepsilon}$ is a data-driven estimate of average annotator disagreement: if the model is 80% confident on the voted class, the remaining 20% is spread uniformly across all K classes. The result is $T^*_{\text{LS}} > T^*_{\text{TS}}$ whenever $\bar{\varepsilon} > 0$, moving the calibrator in the correct direction even without annotator data.

Theoretical interpretation. The smoothing weight $\bar{\varepsilon}$ satisfies a moment-matching fixed point: the expected pseudo-label mass on the voted class equals the average model confidence, $\mathbb{E}[\tilde{\pi}^{\text{LS}}_{y^*}] = \mathbb{E}[\hat{p}_{y^*}]$. Algorithmically, LS-TS can be viewed as a single E-step of an EM algorithm in which the latent annotator distribution is estimated from the current (pre-calibration) model, followed by one M-step that optimises T given those pseudo-labels. This connection to knowledge distillation [Hinton et al., 2015] explains the self-referential nature of the approach: the uncalibrated model both supplies the pseudo-targets and is then calibrated against them. Appendix K compares LS-TS against three simpler smoothing strategies and confirms that the data-driven global $\bar{\varepsilon}$ is the key design choice. Appendix L further shows that even an *adaptive* voted-label calibrator (one that learns a per-instance temperature from logit-derived features) fails to improve over TS, reinforcing that the target distribution, not the flexibility of the calibration map, is the limiting factor.

7 Experiments

We evaluate on four benchmarks with multi-annotator data, each with two backbones: CIFAR-10H (ResNet-50, ViT-B/16), ChaosNLI (RoBERTa-Large, DeBERTa-v3), ISIC 2019 (EfficientNet-B4, ViT-S/16), and DermaMNIST (ResNet-18, ViT-S/16).

Baselines. We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on voted labels [Guo et al., 2017]; (iii) **ATS** (Adaptive Temperature Scaling, Joy et al. 2023): per-instance temperature predicted from four logit-derived features ($\max_k z_k$, prediction entropy, top-2 margin, top-1 confidence) via a linear model, trained with the same voted-label NLL as TS (the most direct adaptive extension of TS without annotator data); (iv) **Platt (PS)**: per-class weight and bias on logits, calibrated against voted labels [Platt, 1999, Kull et al., 2019]; (v) **Dirichlet-Hard**: Dirichlet calibration [Kull et al., 2019] with the full $K \times K$ weight matrix and bias, calibrated against voted labels. Dirichlet-Hard uses the *same architecture* as our Dirichlet-Soft but with voted-label targets; the contrast isolates the effect of the calibration target from model capacity. ATS similarly tests whether richer architecture, without changing the target, is sufficient to fix the calibration gap.

Metrics. We report two families of metrics, which differ in how the ground-truth label is treated:

Binning-based ECE metrics (ECE_{true}, aECE, cwECE) are estimated by drawing $S = 100$ labels $\tilde{y}_i^{(s)} \sim \hat{\pi}(x_i)$ per example and averaging calibration error over draws. ECE_{true} uses $B = 15$ equal-width confidence bins; aECE (adaptive ECE [Nixon et al., 2019]) uses equal-mass bins that adapt to the confidence distribution; cwECE [Kull et al., 2019] averages per-class ECE over K classes. These mirror the evaluation protocol faced by a practitioner who observes one label per prediction at test time.

Soft-target metrics (Brier score Br, NLL) are computed directly against the soft target $\hat{\pi}$:

$$\text{Br}_i = \|\hat{p}(x_i) - \hat{\pi}(x_i)\|^2, \quad \text{NLL}_i = -\sum_k \hat{\pi}_k(x_i) \log \hat{p}_k(x_i) = H\big(\hat{\pi}(x_i),\, \hat{p}(x_i)\big), \tag{5}$$

averaged over the test set. These are equivalent to $\mathbb{E}_{\tilde{y}\sim\hat{\pi}}[\text{Br}(\hat{p}, \tilde{y})]$ and $\mathbb{E}_{\tilde{y}\sim\hat{\pi}}[\text{NLL}(\hat{p}, \tilde{y})]$ respectively, but computed in closed form against the soft target. Brier and NLL are strictly proper scoring rules that directly penalise divergence from the full annotator distribution and thus serve as complementary distributional metrics.

We abbreviate ECE_{true}, aECE, cwECE, Br, NLL throughout.

7.1 CIFAR-10H: Image Classification with Human Label Noise

Dataset and models. CIFAR-10H [Peterson et al., 2019] provides $\approx$51 human annotations per image for the 10 000-image CIFAR-10 test set, directly exposing perceptual ambiguity in low-resolution natural images through repeated human labeling. The test set is split into calibration

Table 1: **CIFAR-10H and ChaosNLI results.** ECE/aECE/cwECE averaged over 100 sampled labels $\tilde{y} \sim \hat{\pi}$; Brier/NLL computed against soft target $\hat{\pi}$ (Eq. 5). Best per architecture in **bold**.

	ResNet-50						**ViT-B/16**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
CIFAR-10H *($\approx$51 annotations per image, K=10)*												
Voted-label baselines												
Uncalibrated	—	4.97	5.71	1.09	0.120	0.692	—	4.99	5.36	1.06	0.111	0.678
TS	2.03	4.29	4.25	0.91	0.112	0.363	2.04	4.48	4.40	0.93	0.106	0.346
ATS	2.33	4.10	4.01	0.87	0.113	0.334	2.26	4.10	4.04	0.86	0.105	0.319
Platt	—	4.29	4.23	0.91	0.112	0.372	—	4.54	4.38	0.93	0.105	0.363
Dirichlet-Hard	—	4.46	4.44	0.95	0.114	0.395	—	4.70	4.62	0.97	0.107	0.394
Ambiguity-aware (ours)												
SLTS	3.18	1.51	1.39	0.45	0.110	0.293	3.07	0.85	0.56	0.39	0.102	0.278
SoftPlatt	—	1.52	1.31	**0.35**	0.110	0.288	—	0.88	0.47	**0.24**	0.101	0.272
VS	—	1.35	1.26	0.39	**0.109**	0.289	—	0.92	0.50	0.32	0.101	0.274
IR-Soft	—	**0.72**	**0.83**	0.45	0.115	0.340	—	0.91	0.61	0.43	0.105	0.321
Dirichlet-Soft	—	1.25	1.06	**0.35**	**0.109**	**0.271**	—	**0.72**	**0.41**	0.25	**0.100**	**0.255**
Annotation-free (ours)												
LS-TS	3.08	1.57	1.31	0.46	0.109	0.293	2.76	2.37	2.21	0.53	0.103	0.285
	RoBERTa-Large						**DeBERTa-v3**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
ChaosNLI *(SNLI+MNLI, 100 annotations per example, K=3)*												
Voted-label baselines												
Uncalibrated	—	27.79	27.77	18.61	0.650	1.384	—	35.97	35.92	24.02	0.743	2.148
TS	2.42	10.55	10.38	7.30	0.537	0.886	3.81	11.63	11.57	8.48	0.549	0.900
ATS	2.28	11.36	11.29	7.73	0.538	0.883	3.57	13.09	13.07	9.06	0.552	0.900
Platt	—	11.16	10.92	7.28	0.530	0.875	—	12.43	12.36	8.43	0.539	0.896
Dirichlet-Hard	—	11.55	11.40	7.64	0.535	0.889	—	12.69	12.63	8.84	0.539	0.895
Ambiguity-aware (ours)												
SLTS	3.41	3.22	2.59	4.62	0.520	0.862	5.28	3.45	3.36	6.53	0.529	0.877
SoftPlatt	—	2.66	2.32	2.43	**0.509**	**0.839**	—	3.17	2.92	3.18	0.511	0.841
VS	—	3.46	3.07	5.13	0.518	0.857	—	3.91	3.35	6.46	0.525	0.869
IR-Soft	—	2.65	2.59	6.24	0.549	0.934	—	**2.15**	3.96	7.20	0.554	0.942
Dirichlet-Soft	—	**2.57**	**1.71**	**2.03**	0.510	**0.839**	—	3.19	**2.71**	**2.95**	**0.509**	**0.836**
Annotation-free (ours)												
LS-TS	4.40	4.16	3.72	5.61	0.522	0.873	6.15	2.65	2.99	6.32	0.529	0.881

(n = 5,000) and evaluation (5,000) by stratified sampling (seed 42). We evaluate two architectures: **ResNet-50** [He et al., 2016] pretrained on ImageNet-1k, fine-tuned for 30 epochs (AdamW, cosine annealing), achieving 97.3% top-1 accuracy. **ViT-B/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol, achieving 98.1% top-1 accuracy.

Results. Both CIFAR-10H backbones show the same pattern: hard-label baselines still leave 4–5% $\mathrm{ECE}_{\mathrm{true}}$, while ambiguity-aware methods reduce it to below 2%. Notably, ATS (adaptive per-instance temperature) achieves 4.10% ECE on both architectures, no better than global TS (4.29% / 4.48%) and far behind LS-TS (1.57% / 2.37%), confirming that per-instance temperature adaptation cannot compensate for the wrong calibration target. Dirichlet-Soft achieves the best Brier score and NLL on both architectures (0.109/0.271 on ResNet-50; 0.100/0.255 on ViT-B/16), confirming that expressive distribution-aware calibrators provide the best distributional fit. LS-TS, using only voted labels, already reduces ECE from 4.29% to 1.57% (ResNet-50) and from 4.48% to 2.37% (ViT-B/16), competitive with fully ambiguity-aware methods.

7.2 ChaosNLI: Natural Language Inference

We next evaluate on **ChaosNLI** [Nie et al., 2020]: 100 human annotations per example for the combined SNLI+MNLI subset (n = 3,113), where disagreement reflects genuine semantic ambiguity rather than annotation noise. We split the data into calibration and test sets (50/50, stratified by majority-vote label to preserve class balance). We evaluate **RoBERTa-Large** [Liu et al., 2019] and

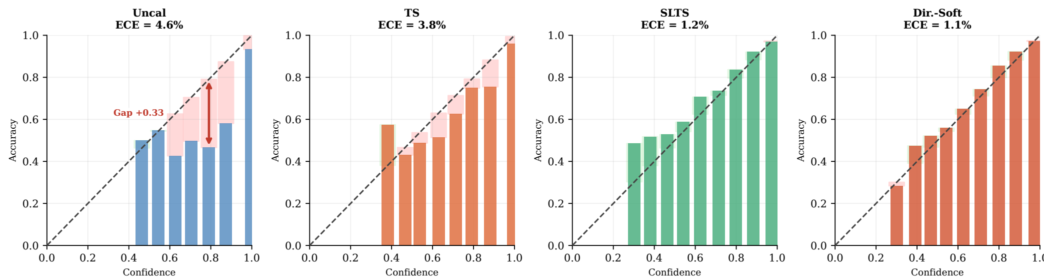


Figure 3: **Reliability diagrams for four representative methods — CIFAR-10H ResNet-50 ($\mathbf{ECE_{true}}$).** Red shading indicates overconfidence; green indicates underconfidence. Uncal and TS remain overconfident; SLTS substantially corrects this with a single temperature parameter; Dirichlet-Soft reduces the residual gap further. Reliability diagrams for all remaining methods are provided in Appendix H.

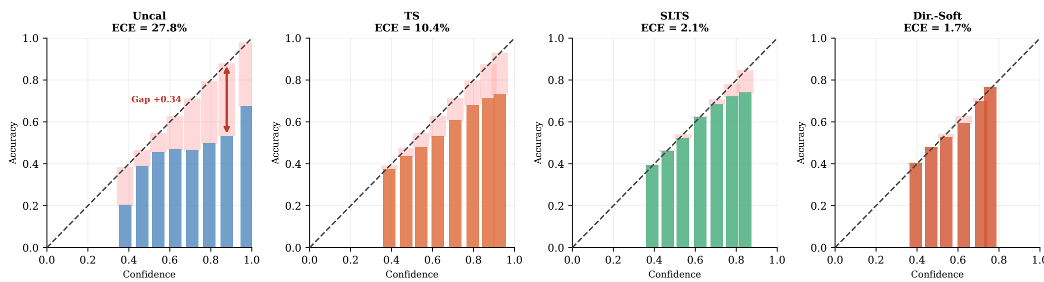


Figure 4: **Reliability diagrams for four representative methods — ChaosNLI RoBERTa-Large ($\mathbf{ECE_{true}}$).** Same method selection as Figure 3. Red shading indicates overconfidence; green indicates underconfidence. NLI models are severely overconfident before calibration; ambiguity-aware methods substantially correct this.

DeBERTa-v3-base [He et al., 2023], both fine-tuned on MNLI training data; ChaosNLI draws from the SNLI and MNLI dev/test splits, so no training examples are re-used at evaluation.

Results. The same pattern holds in NLI: hard-label baselines leave 10–12% ECE, and ATS (11.36% on RoBERTa-L, 13.09% on DeBERTa-v3) performs *worse* than global TS on DeBERTa-v3, further demonstrating that adaptive capacity without the correct target is ineffective. The best distributional fit is achieved by Dirichlet-Soft and SoftPlatt: on RoBERTa-L, SoftPlatt achieves the best Brier (0.509 vs. Dir-Soft 0.510) and both tie on NLL (0.839); on DeBERTa-v3, Dirichlet-Soft achieves the best Brier (0.509) and NLL (0.836). LS-TS reduces ECE from 10.55% to 4.16% (RoBERTa-L) and from 11.63% to 2.65% (DeBERTa-v3) using only voted labels, outperforming all hard-label baselines by a wide margin.

7.3 ISIC 2019: Skin Disease Diagnosis

We evaluate on **ISIC 2019** [Tschandl et al., 2018, Combalia et al., 2019] (8 skin conditions, $\approx$25,000 images), a clinically realistic differential-diagnosis task in which several lesion categories have overlapping visual morphology. We use $m = 9$ synthetic dermatologist annotations per image, generated by sampling from a clinically calibrated 8×8 confusion matrix following **?** (Appendix I), with diagonal agreement matched to Liu et al. [2020]: mean 75%. Because no per-image reader study is publicly available for this dataset, annotation ambiguity is modelled *class-conditionally*: each image's confusion depends on its consensus class, not on its individual visual content, which is a deliberate simplification of true instance-level annotator variation. We evaluate **EfficientNet-B4** [Tan and Le, 2019] and **ViT-S/16** [Dosovitskiy et al., 2021] on a stratified 70/15/15 split.

ISIC 2019 results. The medical setting, where annotation ambiguity is modelled synthetically from clinician confusion rates (class-conditional, not instance-level), shows the same failure mode for

Table 2: **ISIC 2019 and DermaMNIST results.** ECE/aECE/cwECE averaged over 100 sampled labels $\tilde{y} \sim \hat{\pi}$; Brier/NLL computed against soft target $\hat{\pi}$ (Eq. 5). Best per architecture in **bold**.

	EfficientNet-B4						**ViT-S/16**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
ISIC 2019 *(m=9 synthetic annotators, mean diagonal agreement* 75%*)*												
Voted-label baselines												
Uncalibrated	—	25.76	25.84	6.67	0.600	2.710	—	23.31	23.33	6.55	0.651	1.978
TS	1.79	18.54	18.49	5.02	0.560	1.653	1.42	16.59	16.60	5.23	0.617	1.563
ATS	1.71	18.83	18.80	5.03	0.560	1.733	1.12	17.53	17.48	5.46	0.631	1.730
Platt	—	19.89	19.81	5.24	0.562	1.659	—	17.19	17.17	4.74	0.597	1.583
Dirichlet-Hard	—	20.36	20.32	5.33	0.563	1.674	—	17.76	17.72	4.86	0.600	1.597
Ambiguity-aware (ours)												
SLTS	4.46	9.71	9.74	2.92	0.528	1.146	3.15	6.96	6.68	3.29	0.586	1.233
SoftPlatt	—	9.25	9.31	2.23	0.523	1.108	—	8.05	7.90	2.05	0.565	1.190
VS	—	9.03	9.11	2.42	0.524	1.121	—	8.50	8.59	2.24	0.567	1.198
IR-Soft	—	**1.75**	**2.43**	4.28	0.538	1.284	—	**1.73**	**1.59**	5.78	0.621	1.466
Dirichlet-Soft	—	7.50	7.48	**2.00**	**0.518**	**1.084**	—	6.85	6.80	**1.77**	**0.563**	**1.175**
Annotation-free (ours)												
LS-TS	4.30	9.34	9.46	2.98	0.528	1.149	3.98	15.49	15.47	4.52	0.619	1.303
	ResNet-18						**ViT-S/16**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
DermaMNIST *(m=5 synthetic annotators, overall agreement* 64.7%*)*												
Voted-label baselines												
Uncalibrated	—	34.35	34.26	10.15	0.767	2.689	—	37.12	37.10	10.82	0.786	3.766
TS	1.86	23.05	22.82	6.97	0.686	1.654	2.60	24.36	24.26	7.29	0.683	1.664
ATS	1.86	22.82	22.75	6.97	0.687	1.606	2.53	25.35	25.24	7.53	0.686	1.650
Platt	—	23.64	23.56	7.08	0.682	1.667	—	23.82	23.59	7.10	0.675	1.681
Dirichlet-Hard	—	24.58	24.46	7.33	0.690	1.771	—	24.84	24.68	7.49	0.682	1.897
Ambiguity-aware (ours)												
SLTS	3.51	4.61	4.44	3.56	0.622	1.376	5.11	3.58	3.06	3.37	0.610	1.350
SoftPlatt	—	4.19	3.55	1.61	0.612	1.312	—	3.56	2.93	1.29	**0.601**	1.289
VS	—	4.45	3.83	3.23	0.621	1.363	—	4.03	3.71	2.84	0.611	1.331
IR-Soft	—	**2.15**	**2.34**	4.55	0.634	1.440	—	3.79	3.73	4.59	0.628	1.431
Dirichlet-Soft	—	3.55	3.36	**1.22**	**0.610**	**1.305**	—	**3.32**	**2.87**	**1.14**	**0.601**	**1.286**
Annotation-free (ours)												
LS-TS	3.03	5.05	4.82	3.79	0.630	1.394	4.19	7.36	7.06	3.68	0.615	1.362

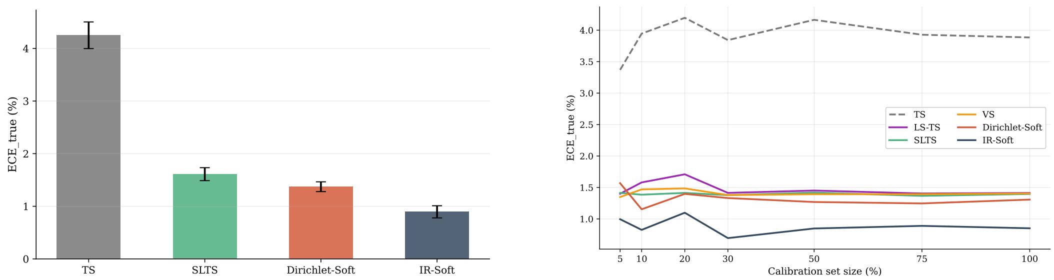
standard calibrators and the same benefit from ambiguity-aware targets. Dirichlet-Soft again achieves the best Brier and NLL on both backbones (0.518/1.084 on ENet-B4; 0.563/1.175 on ViT-S/16), while IR-Soft achieves the lowest ECE. LS-TS halves the ECE of TS on ENet-B4 (18.54%→9.34%) using only voted labels; the improvement is smaller on ViT-S/16 (16.59%→15.49%), highlighting that model confidence is not always a reliable proxy for annotation ambiguity.

7.4 DermaMNIST: Supplementary Skin Disease Experiment

We additionally evaluate on **DermaMNIST** [Yang et al., 2023] (7-class, derived from HAM10000 [Tschandl et al., 2018]; $m = 5$ synthetic annotators; overall agreement $\approx 64.7\%$) with two backbones: ResNet-18 and ViT-S/16. Results in Table 2 show that TS leaves ECE $= 23.1\%$ for ResNet-18 while ambiguity-aware methods reduce it substantially: SLTS achieves 4.6% (80% reduction), IR-Soft achieves 2.2% (91% reduction), and Dirichlet-Soft achieves 3.6% while attaining the best cwECE (1.2%), Brier (0.610), and NLL (1.305). ViT-S/16 shows a similar pattern (SLTS: 3.6%; Dirichlet-Soft achieves the best ECE, Brier, and NLL: 3.3%/0.601/1.286). Both backbones show $T_{\text{SLTS}} > T_{\text{TS}} > 1$: SLTS requires substantially higher temperature to match the more diffuse annotation distribution.

7.5 Reliability Diagrams

Figure 3 shows reliability diagrams for four representative methods on CIFAR-10H (ResNet-50). The diagrams are evaluated against the true label distribution (ECE_{true}). Uncal and TS are systematically



(a) Multi-seed stability. Mean $\mathrm{ECE}_{\mathrm{true}}$ ($\pm$std) over 5 seeds; ranking is stable across splits.

(b) Calibration-set-size ablation. $\mathrm{ECE}_{\mathrm{true}}$ vs. fraction of calibration data; ambiguity-aware methods converge at 5–10%.

Figure 5: **Robustness of ambiguity-aware calibration — CIFAR-10H ResNet-50.** Full results for all architectures in Appendices G and E; annotation-count ablation in Appendix F.

overconfident: bars fall well below the diagonal at high confidence, leaving a large red gap. SLTS corrects the overconfidence with a scalar temperature adjustment, aligning most bins close to the diagonal. Dirichlet-Soft achieves near-perfect calibration by combining the expressive $K \times K$ affine transform with the correct distributional target. Figure 4 shows corresponding diagrams for ChaosNLI (RoBERTa-Large); full 3×5 panels covering all 13 methods and all remaining architectures are provided in Appendix H.

8 Discussion

Sacrificing voted-label calibration. A natural question is whether ambiguity-aware methods degrade $\mathrm{ECE}_{\mathrm{voted}}$. This is expected behaviour: voted-label calibration is epistemically unreliable when annotators genuinely disagree, as the voted label overstates the majority-class probability. Practitioners who must satisfy a hard $\mathrm{ECE}_{\mathrm{voted}}$ constraint can apply TS on top of Dirichlet-Soft or SLTS outputs.

Practical guidelines. Use **Dirichlet-Soft** as the default when annotator distributions are available: it achieves the best or near-best Brier score and NLL on most benchmarks (best NLL in 7/8 settings; best or tied Brier in 7/8 settings) while maintaining competitive ECE; SoftPlatt is competitive on Brier for NLI tasks. Use **SLTS** when a single-parameter method is preferred for simplicity; **IR-Soft** when the lowest ECE is the sole criterion; and **LS-TS** when the calibration set contains only one-hot voted labels.

Robustness. Figure 5a confirms that the ranking of methods is stable across random seeds (std $\leq$ 0.54 pp); full results for all architectures are in Appendix G. Figure 5b shows that ambiguity-aware methods converge with as few as 5–10% of the calibration set; full results for all architectures are in Appendix E.

Limitations. Full ambiguity-aware calibration (including Dirichlet-Soft) requires multi-annotator labels at calibration time. When only voted labels are available, LS-TS remains useful but does not fully match Dirichlet-Soft, particularly on datasets where model confidence diverges from annotation ambiguity (e.g., ISIC 2019 ViT-S/16). ISIC 2019 and DermaMNIST rely on synthetic annotators derived from clinician-reported agreement. Crucially, this is a *class-conditional* model: annotation confusion depends only on the consensus class label, not on instance-level visual difficulty. Real multi-reader medical datasets with per-image reader distributions would provide stronger validation and would capture the instance-level variation our synthetic model does not; concrete candidates include VinDr-CXR [Nguyen et al., 2022] (3 independent radiologists per chest X-ray, 15k images) and CheXpert [Irvin et al., 2019] (5 independent radiologist annotations on the test set), both of which provide true per-image annotator distributions directly amenable to our framework. Conclusions about absolute ECE magnitudes on these two datasets should be interpreted with this in mind; the relative ranking of methods is expected to be robust to the specific confusion matrix used, as confirmed by the multi-seed annotation-generation ablation (Appendix G).

9 Conclusion

We formalized the problem of confidence calibration under ambiguous ground truth and showed that standard post-hoc calibrators fitted on voted labels are systematically miscalibrated against the underlying ambiguous label distribution. Our flagship ambiguity-aware method, **Dirichlet-Soft**, requires no retraining and achieves the best or near-best Brier score and NLL across most benchmarks and architectures (best NLL in 7 of 8 settings; best or tied Brier in 7 of 8 settings). On CIFAR-10H and ChaosNLI (benchmarks with real per-instance multi-annotator distributions), it reduces true-label ECE by 59–86% relative to Temperature Scaling; ISIC 2019 and DermaMNIST results (with synthetic annotators) provide supportive evidence across medical imaging tasks. The key insight is that the calibration target, not method capacity, is the limiting factor, evidenced by three complementary ablations: Dirichlet calibration with voted-label targets (Dirichlet-Hard) consistently *underperforms* simple TS; per-instance temperature adaptation (ATS) trained with voted-label supervision yields no improvement over TS and sometimes degrades it; and the same global-temperature architecture as TS, when supplied with smoothed pseudo-targets (LS-TS), reduces ECE by 7–78%. Together these confirm that architectural flexibility without the right supervision signal is insufficient, while even a simple shift in the calibration target produces reliable gains. Crucially, when the calibration set contains only one-hot voted labels, our annotation-free method **LS-TS** still improves true-label calibration, reducing ECE by 7–78% using only the model's own confidence as a proxy for annotator disagreement. We recommend Dirichlet-Soft as the default wherever multi-annotator data is available, and LS-TS as a practical post-hoc alternative when only voted labels are available.

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A Dataset Background and Sources of Label Ambiguity

All four benchmarks in this paper are appropriate for studying calibration under ambiguous ground truth, but they exhibit ambiguity in slightly different ways.

CIFAR-10H. Peterson et al. [2019] introduced CIFAR-10H by collecting a full distribution of human labels for each image in the CIFAR-10 test set, explicitly to capture *human perceptual uncertainty*. The ambiguous examples are typically low-resolution images whose visual evidence supports multiple plausible classes for human observers. In this benchmark, the label distribution is therefore directly observed from repeated human annotation, making it a clean testbed for ambiguity-aware calibration.

ChaosNLI. Nie et al. [2020] introduced ChaosNLI to study *collective human opinions* on natural language inference. Each example receives 100 human labels, and the paper reports that substantial disagreement is common rather than exceptional. This ambiguity is semantic rather than perceptual: differences in pragmatic assumptions, underspecification, and sentence interpretation lead reasonable annotators to assign different NLI labels to the same premise–hypothesis pair.

ISIC 2019. ISIC 2019 is a multiclass dermoscopic lesion diagnosis benchmark assembled from several clinical sources, including HAM10000 [Tschandl et al., 2018] and BCN20000 [Combalia et al., 2019]. The task itself is ambiguity-prone because several lesion categories share overlapping visual morphology, and the challenge was designed to reflect the more realistic task of differential diagnosis rather than a simpler benign-versus-malignant decision. Because the public challenge release does not provide per-image reader distributions, we model ambiguity using synthetic annotators calibrated to dermatologist confusion patterns and reader agreement reported in Liu et al. [2020]. This preserves the core property relevant to our study: in dermatology, a single image can support multiple clinically plausible labels.

DermaMNIST. DermaMNIST [Yang et al., 2023] is a standardized 28×28 version of the dermatology benchmark derived from HAM10000 [Tschandl et al., 2018]. HAM10000 consists of dermatoscopic images of common pigmented lesions collected in clinical practice, where diagnosis already involves fine-grained distinctions among visually similar categories such as melanoma, nevi, and benign keratoses. DermaMNIST inherits this medical ambiguity from HAM10000 and, because it is aggressively downsampled for lightweight benchmarking, removes additional visual detail that can further increase label uncertainty. As with ISIC 2019, we therefore evaluate calibration against an ambiguity-aware label distribution rather than treating the majority diagnosis as uniquely correct.

B Proof of Proposition 1

Let $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c,\, x_i \in \mathcal{C}\}$ be the subset of calibration examples from the ambiguous cluster $\mathcal{C}$, where the voted label is always class c. Denote $p_i = \operatorname{softmax}(z_i)_c$.

The TS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \operatorname{softmax}(z_i/T)_c$. As $T \to 0^+$, $\operatorname{softmax}(z_i/T)_c \to 1$ (assuming $z_{ic} > z_{ik}$ for all $k \neq c$), so $\mathcal{L}_{\text{TS}} \to 0$. Hence $\partial \mathcal{L}_{\text{TS}}/\partial T > 0$ for all $T > 0$: decreasing T always decreases the voted-label loss on the ambiguous cluster. The globally optimal T^*_{TS} balances this downward pull from ambiguous examples against the upward pull from unambiguous examples, but $T^*_{\text{TS}} \leq T^*_{\text{no-amb}}$ (the optimal temperature without the ambiguous cluster): including ambiguous examples drives the optimum downward.

The soft-label loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{soft}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \operatorname{softmax}(z_i/T)_k$, with $\hat{\pi}_c(x_i) = q < 1$. The minimum-loss T^*_{soft} satisfies $\operatorname{softmax}(z_i/T^*_{\text{soft}})_c \approx q$, requiring $T^*_{\text{soft}} > T^*_{\text{TS}}$ whenever $p_i > q$. Hence $T^*_{\text{TS}} < T^*_{\text{soft}}$ regardless of whether either optimum exceeds 1. In the special case where the model is already overconfident on the ambiguous cluster ($p_i \gg q$), this gap is large enough that $T^*_{\text{TS}} < 1$; in practice both optima exceed 1 (since the model is also underconfident on unambiguous examples), but the inequality $T^*_{\text{TS}} < T^*_{\text{soft}}$ is observed consistently across all experimental settings. $\square$

C Proof of Proposition 2

We formalize the argument given in the proof sketch in the main text.

Setup. Let $p = \hat{p}_{\hat{c}}(x)$ denote the model's predicted confidence for the top class $\hat{c}$ and let $q = \pi_{\hat{c}}(x) \in (0, 1]$ denote the true annotator probability for that class. We assume the model is trained against voted labels, so p is approximately constant across examples with the same voted label regardless of their annotation entropy $H(x)$.

Voted-label calibration error contribution. For example x_i in confidence bin B_b (where $\overline{p}(B_b) \approx p$), the per-example voted-label calibration error is $|p - \mathbf{1}[\hat{c} = y^*]|$. Since $\hat{c} = y^*$ for correctly-classified ambiguous examples (the model predicts the majority class), this contribution is approximately $|p - 1|$ for examples where the model makes the right hard prediction, independent of $H(x)$.

True-label calibration error contribution. The per-example true-label calibration error is $|p - \pi_{\hat{c}}(x)| = |p - q|$.

Excess true-label error relative to voted-label evaluation. The per-example excess contribution is:

$$\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|. \tag{6}$$

When the model is correctly calibrated to voted labels ($p \approx \mathbf{1}[\hat{c} = y^*]$ in expectation), the voted contribution is near zero: $|p - 1| \approx 0$ for high-accuracy bins and $|p - 0| = p$ for error bins. For unambiguous examples ($H(x) = 0$), $q = \pi_{y^*} = 1$, so $\delta(x) = |p - 1| - |p - 1| = 0$.

Monotonicity. As $H(x)$ increases, $q = \pi_{y^*}(x)$ decreases below 1 (more probability mass shifts to non-majority classes). Specifically, for a K-class problem with uniform annotator distribution at maximum entropy, $q = 1/K$. The true-label contribution $|p - q|$ increases strictly as q decreases from 1 (since $p > q$ for a well-trained model predicting the majority class with confidence exceeding the majority probability). The voted-label contribution $|p - \mathbf{1}[\hat{c} = y^*]|$ is unaffected by $H(x)$. Therefore $\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|$ is non-decreasing in $H(x)$, and strictly increasing whenever $p > q > 0$.

Aggregate monotonicity. The aggregate excess true-label error relative to voted-label evaluation equals $\sum_b \frac{|B_b|}{n} \sum_{i \in B_b} \delta(x_i)/|B_b|$. Since $\delta(x_i)$ is non-decreasing in $H(x_i)$ and bins contain a mixture of examples, this discrepancy grows with the mean annotation entropy of the test set. □

Remark 1 (Independence assumption)**.** *Assumption (i) (that $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$) may be violated in practice if harder examples (high $H(x)$) also have lower model confidence. When the two are positively correlated, the per-example discrepancy term $\delta(x)$ grows even faster with $H(x)$ than the proof suggests, so Proposition 2 remains valid and the monotonicity is if anything* conservative.

D Monte Carlo Temperature Scaling (MCTS)

Method. When individual annotation records are available rather than pre-aggregated label distributions, **MCTS** draws S samples $a_{is} \sim \hat{\pi}(x_i)$ per example and minimizes

$$\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS} \sum_{i=1}^{n} \sum_{s=1}^{S} \log \operatorname{softmax}(z_i/T)_{a_{is}}.$$

MCTS is a strictly more general formulation than SLTS: it applies even when only raw annotation samples are stored rather than pre-computed frequency vectors.

A single annotation suffices. The key practical implication of MCTS is that *even* $S = 1$ (a single randomly drawn annotation per calibration example) already achieves ECE_{true} of 1.53%, matching SLTS to within rounding error and reducing TS's 4.29% by 64%. This means that any dataset where each example has been annotated even once by a randomly selected annotator (rather than deterministically assigned the majority label) is immediately amenable to strong ambiguity-aware calibration. Formally, since $\mathbb{E}_{\hat{y}\sim\hat{\pi}}[\text{CE}(\operatorname{softmax}(z/T), \hat{y})] = \text{KL}(\hat{\pi} \| \operatorname{softmax}(z/T)) + H(\hat{\pi})$, the MCTS objective converges to SLTS in expectation as $S \to \infty$, with variance $\propto 1/S$. Table 3 confirms the fast convergence: the ECE gap between $S = 1$ and $S \to \infty$ is only 0.02 pp; in practice S = 20–50 is sufficient to reduce variance to negligible levels.

Table 3: **MCTS convergence on CIFAR-10H (ResNet-50).** Mean ± std over 5 seeds; S = MC annotation samples per calibration example.

Method	ECE (%)	T^*
TS (voted labels)	4.29	2.030
MCTS $S = 1$	1.53 ± 0.01	3.174 ± 0.041
MCTS $S = 5$	1.53 ± 0.02	3.166 ± 0.014
MCTS $S = 20$	1.52 ± 0.00	3.187 ± 0.013
MCTS $S = 50$	1.52 ± 0.01	3.185 ± 0.007
MCTS $S = 200$	1.52 ± 0.00	3.178 ± 0.003
SLTS ($S \to \infty$)	1.51	3.180

E Ablation: Calibration Set Size

We fix the annotation count at the full observed value and vary the fraction of calibration examples used, from 5% to 100% (stratified subsampling by class). Results are shown for all four benchmarks and eight architectures.

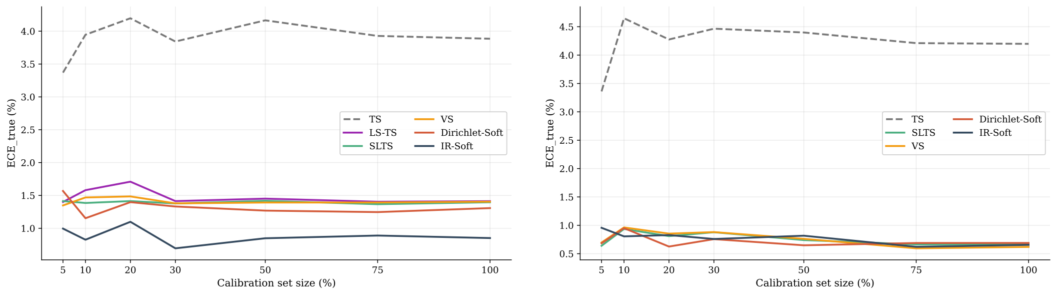

Figure 6: **Calibration-set-size ablation — CIFAR-10H.** Left: ResNet-50. Right: ViT-B/16. TS's ECE is flat across all calibration set sizes; ambiguity-aware methods converge within 5–10%.

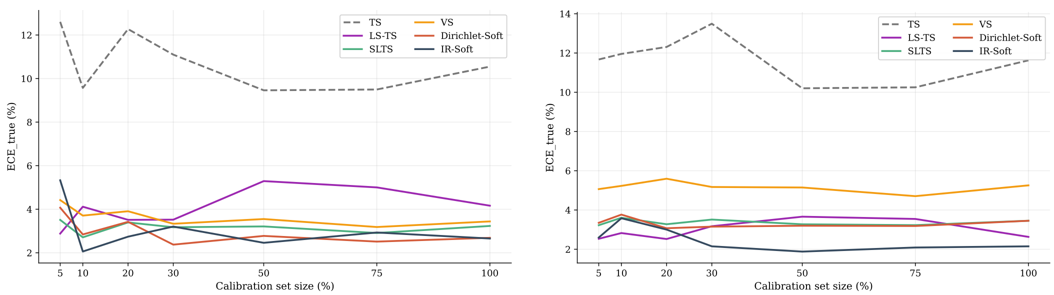

Figure 7: **Calibration-set-size ablation — ChaosNLI.** Left: RoBERTa-Large. Right: DeBERTa-v3-base. NLI models show the same pattern: TS is flat near 10–13% while soft methods stabilise near 3–4%.

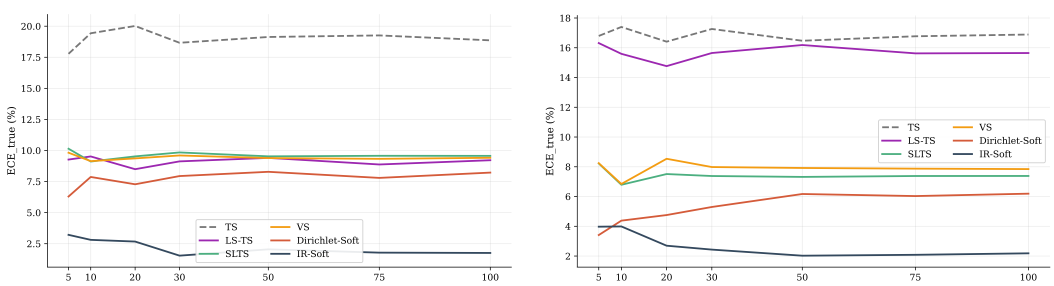

Figure 8: **Calibration-set-size ablation — ISIC 2019.** Left: EfficientNet-B4. Right: ViT-S/16. IR-Soft achieves the lowest ECE across all calibration set sizes on both architectures.

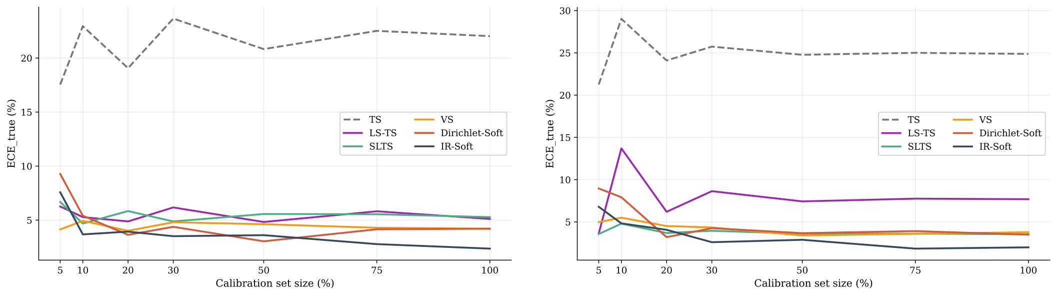

Figure 9: **Calibration-set-size ablation — DermaMNIST.** Left: ResNet-18. Right: ViT-S/16. Ambiguity-aware methods converge quickly; TS remains flat regardless of calibration set size.

Key findings. Across all four benchmarks and eight architectures, TS's ECE does not decrease with more calibration data, confirming that the degradation is a target bias rather than a variance artifact. In contrast, SLTS achieves stable low ECE from as few as 5–10% of the calibration set, and IR-Soft and Dirichlet-Soft converge even faster on most benchmarks. This suggests that the annotation distribution (not the number of calibration examples) is the limiting factor for ambiguity-aware methods.

F Ablation: Number of Annotations per Image

We study how performance depends on the number of annotations m per calibration image. We fix the calibration set size and the annotation generation seed, and sweep m from 1 to 50, resampling the soft targets for each value of m. Results are shown for CIFAR-10H, which provides up to ≈51 real annotations per image; we subsample from the full pool.

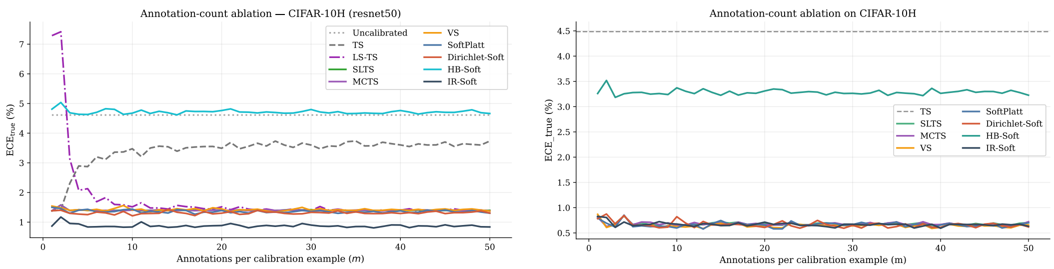

Figure 10: **Annotation-count ablation — CIFAR-10H.** $\mathrm{ECE}_{\mathrm{true}}$ vs. annotations per image m for ResNet-50 (left) and ViT-B/16 (right). Ambiguity-aware methods are stable for $m \geq 2$; TS rises with m due to target mismatch; LS-TS converges by $m \approx 4$.

Key findings. Ambiguity-aware calibration is robust to annotation count once $m \geq 2$: SLTS, Dirichlet-Soft, and IR-Soft achieve near-minimal ECE with only a handful of annotations per image and do not improve substantially beyond $m = 5$. LS-TS converges quickly (by $m \approx 4$) because the global smoothing parameter ε stabilises once each image has at least a few annotations. Standard Temperature Scaling shows the opposite trend: its $\mathrm{ECE}_{\mathrm{true}}$ *increases* with m because a higher temperature is required to match the more diffuse annotation distribution, exposing the fundamental target mismatch.

G Statistical Significance: Multi-Seed Analysis

For CIFAR-10H and ChaosNLI we repeat the evaluation with five random seeds (42–46), each producing a different stratified 50/50 cal/test split. For ISIC 2019 and DermaMNIST the val/test split is predetermined, so we instead vary the annotation generation seed (42–46), testing sensitivity to the particular synthetic annotation sample. Table 4 reports mean $\pm$ std of ECE_{true} across seeds for all eight (dataset, arch) combinations.

Table 4: **Multi-seed stability — all benchmarks and architectures.** Mean $\pm$ std of ECE_{true} (%) over 5 seeds. For CIFAR-10H and ChaosNLI the seed varies the cal/test split; for ISIC 2019 and DermaMNIST it varies the synthetic annotation sample. Standard deviations are small relative to the ECE gaps in all cases.

Dataset	Arch	TS	SLTS	Dir.-Soft	IR-Soft
CIFAR-10H	ResNet-50	4.25 ± 0.25	1.61 ± 0.12	1.37 ± 0.09	0.89 ± 0.12
CIFAR-10H	ViT-B/16	4.39 ± 0.20	0.83 ± 0.14	0.74 ± 0.07	0.88 ± 0.16
ChaosNLI	RoBERTa-L	11.54 ± 0.54	3.85 ± 0.34	2.90 ± 0.14	2.40 ± 0.18
ChaosNLI	DeBERTa-v3	12.73 ± 0.60	4.41 ± 0.54	3.80 ± 0.35	2.61 ± 0.41
ISIC 2019	ENet-B4	18.84 ± 0.17	9.80 ± 0.22	8.45 ± 0.20	1.79 ± 0.16
ISIC 2019	ViT-S/16	16.93 ± 0.16	7.30 ± 0.13	6.20 ± 0.19	2.05 ± 0.13
DermaMNIST	ResNet-18	22.43 ± 0.28	5.08 ± 0.25	3.53 ± 0.39	2.33 ± 0.17
DermaMNIST	ViT-S/16	24.81 ± 0.08	3.61 ± 0.28	3.11 ± 0.21	2.09 ± 0.14

Across all benchmarks and architectures, standard deviations are small relative to the ECE gaps: TS vs. ambiguity-aware methods differ by 2–10$\times$ the standard deviation, confirming statistical robustness of the results in Tables 1–2.

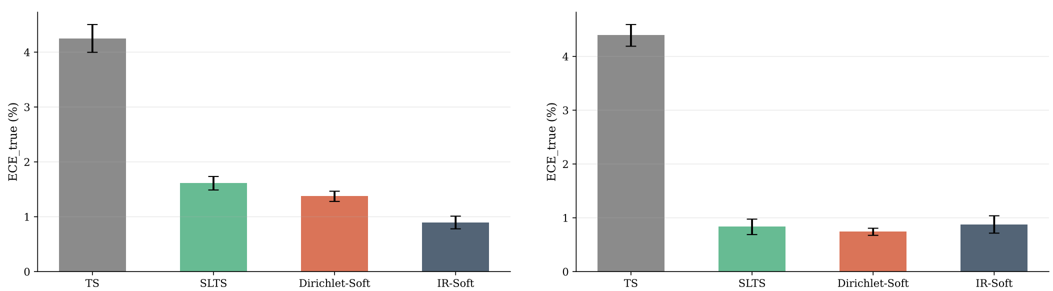

Figure 11: **Multi-seed bar charts — CIFAR-10H.** Left: ResNet-50. Right: ViT-B/16.

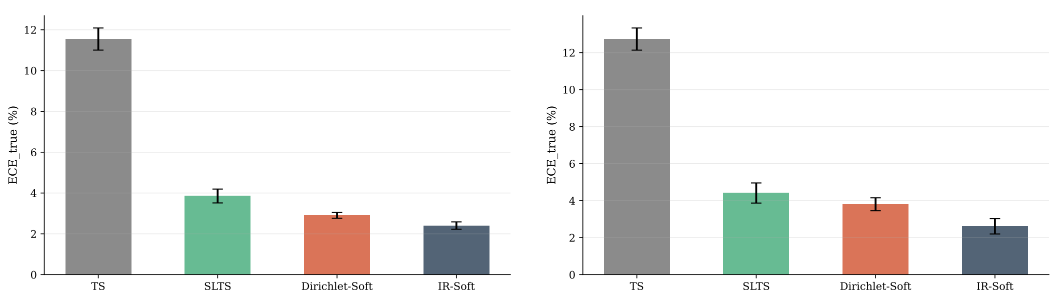

Figure 12: **Multi-seed bar charts — ChaosNLI.** Left: RoBERTa-Large. Right: DeBERTa-v3-base.

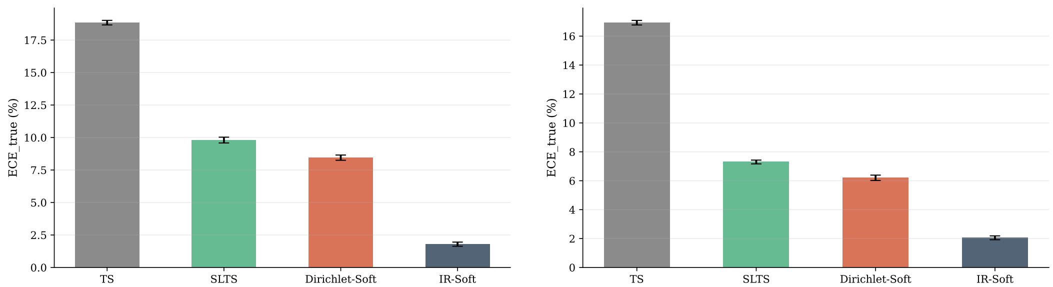

Figure 13: **Multi-seed bar charts — ISIC 2019.** Left: EfficientNet-B4. Right: ViT-S/16. Seed varies the annotation generation; val/test split is fixed.

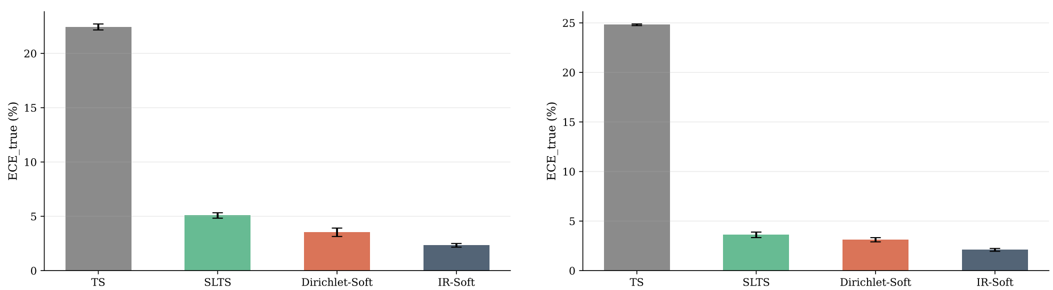

Figure 14: **Multi-seed bar charts — DermaMNIST.** Left: ResNet-18. Right: ViT-S/16. Seed varies the annotation generation; val/test split is fixed.

H Reliability Diagrams

Four-method summaries for CIFAR-10H (ResNet-50) and ChaosNLI (RoBERTa-Large) are shown in Figures 3 and 4 in the main text. Here we provide the four-method summary panels and full panels for all remaining architectures and datasets. All panels share the same layout: top row = voted-label baselines (Uncal, TS, Platt, Dir.-Hard); middle row = annotation-free (LS-TS) and temperature-based soft methods (SLTS, MCTS, SoftPlatt, VS); bottom row = non-parametric/matrix soft methods (Dir.-Soft, IR-Soft). Red shading = overconfident; green = underconfident.

CIFAR-10H

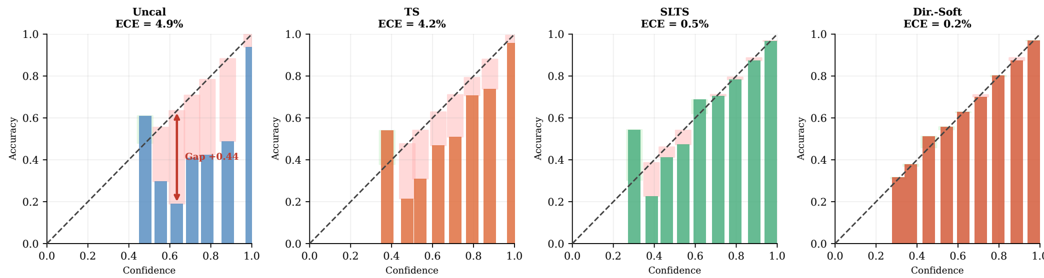

Figure 15: **Reliability diagrams (summary) — CIFAR-10H ViT-B/16.**

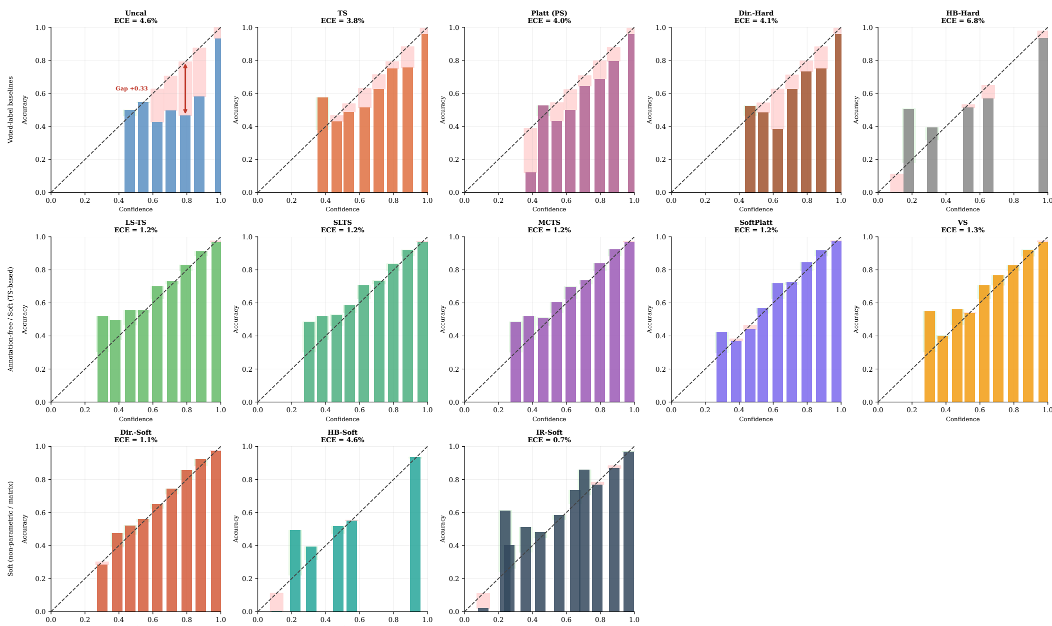

Figure 16: **Reliability diagrams (all methods) — CIFAR-10H ResNet-50 ($\mathbf{ECE_{true}}$).**

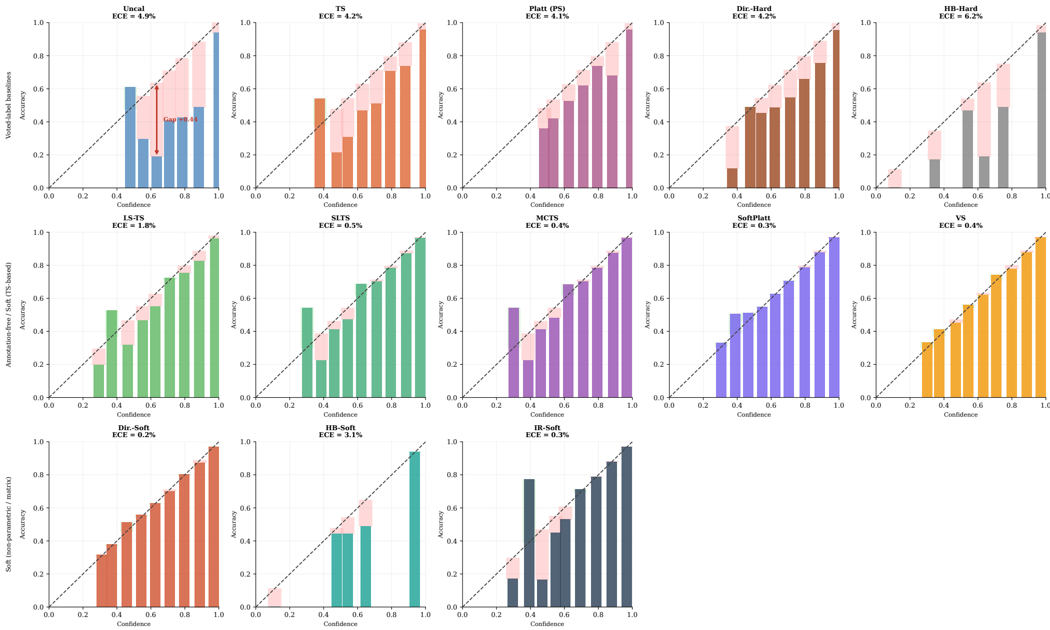

Figure 17: **Reliability diagrams (all methods) — CIFAR-10H ViT-B/16 (ECE_{true}).**

ChaosNLI

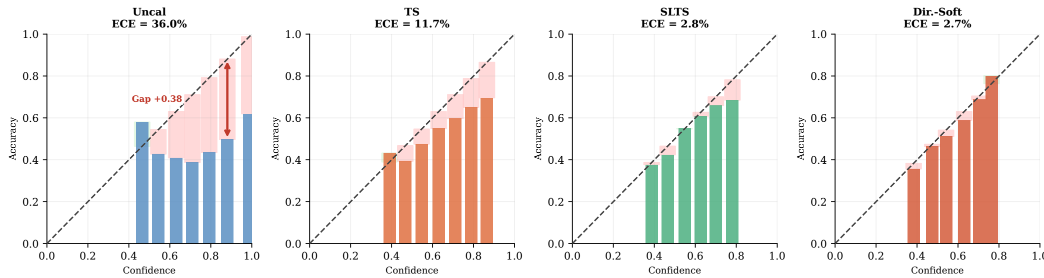

Figure 18: **Reliability diagrams (summary) — ChaosNLI DeBERTa-v3-base.**

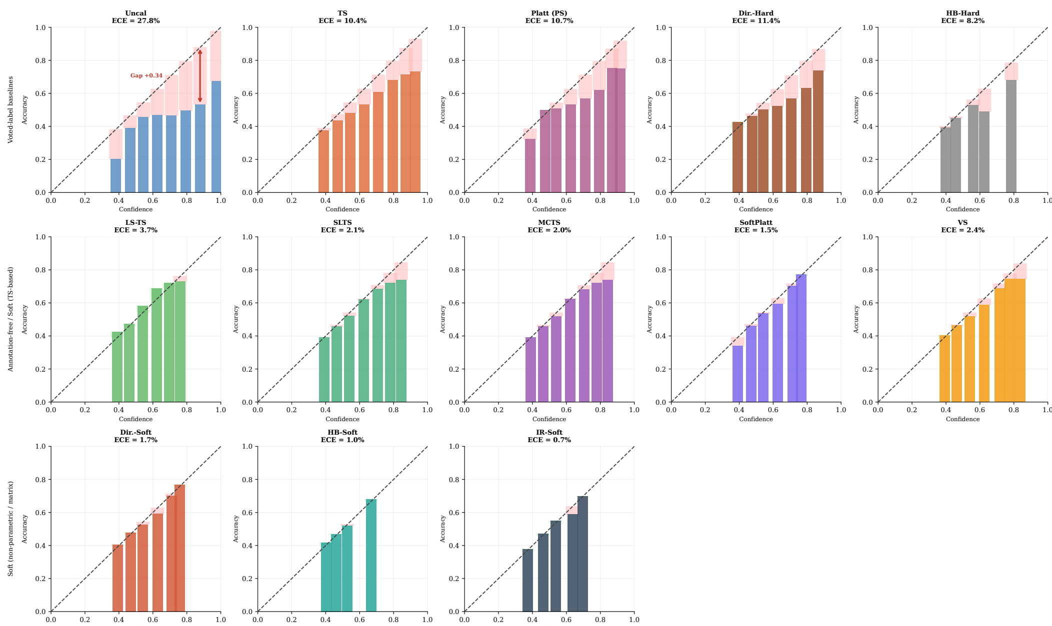

Figure 19: **Reliability diagrams (all methods) — ChaosNLI RoBERTa-Large (ECE$_{\text{true}}$).** NLI models are severely overconfident before calibration; ambiguity-aware methods substantially correct this.

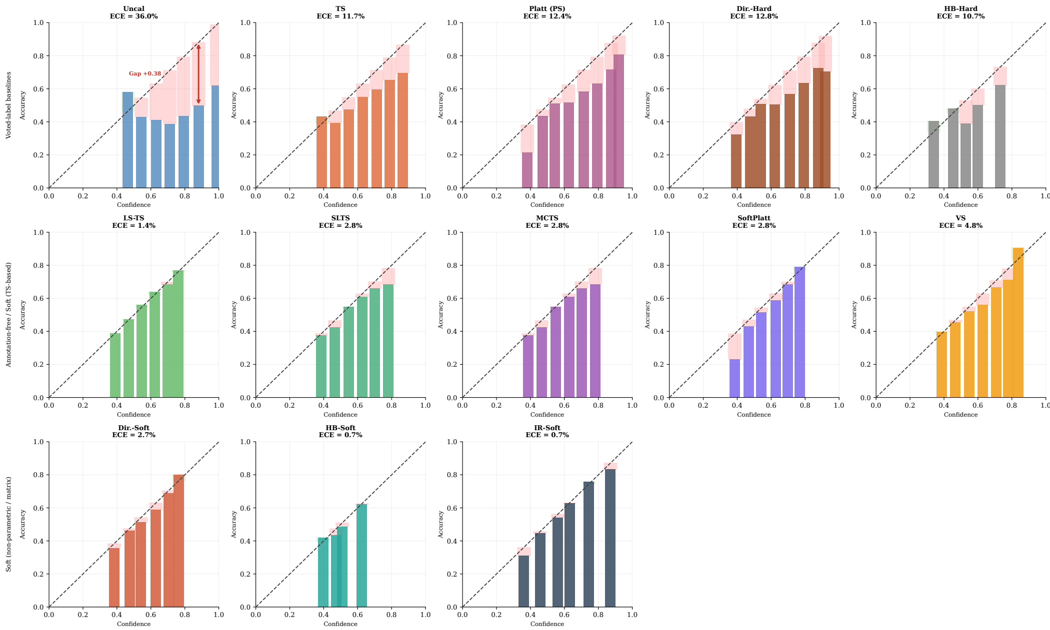

Figure 20: **Reliability diagrams (all methods) — ChaosNLI DeBERTa-v3-base ($\mathbf{ECE_{true}}$).**

ISIC 2019

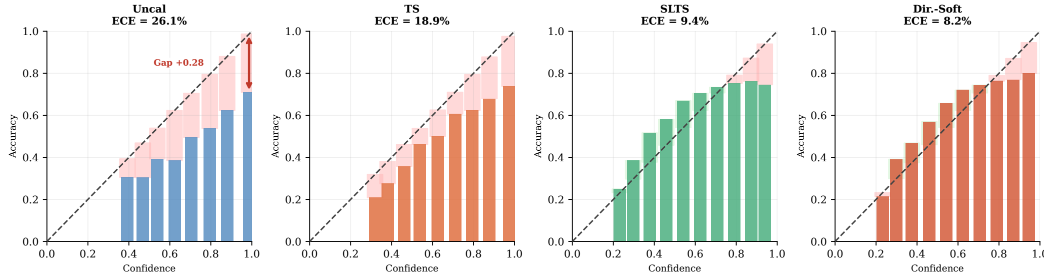

Figure 21: **Reliability diagrams (summary) — ISIC 2019 EfficientNet-B4.**

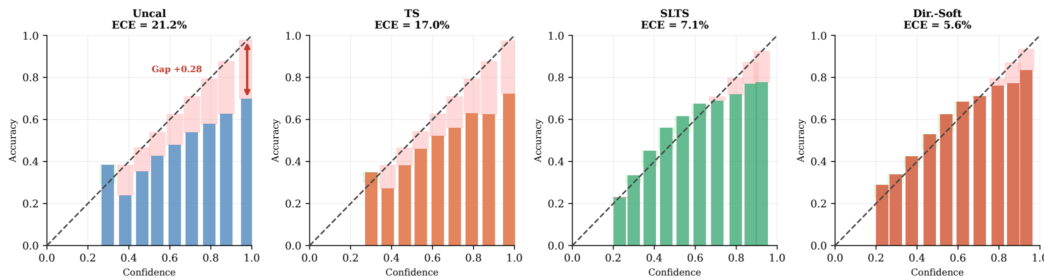

Figure 22: **Reliability diagrams (summary) — ISIC 2019 ViT-S/16.**

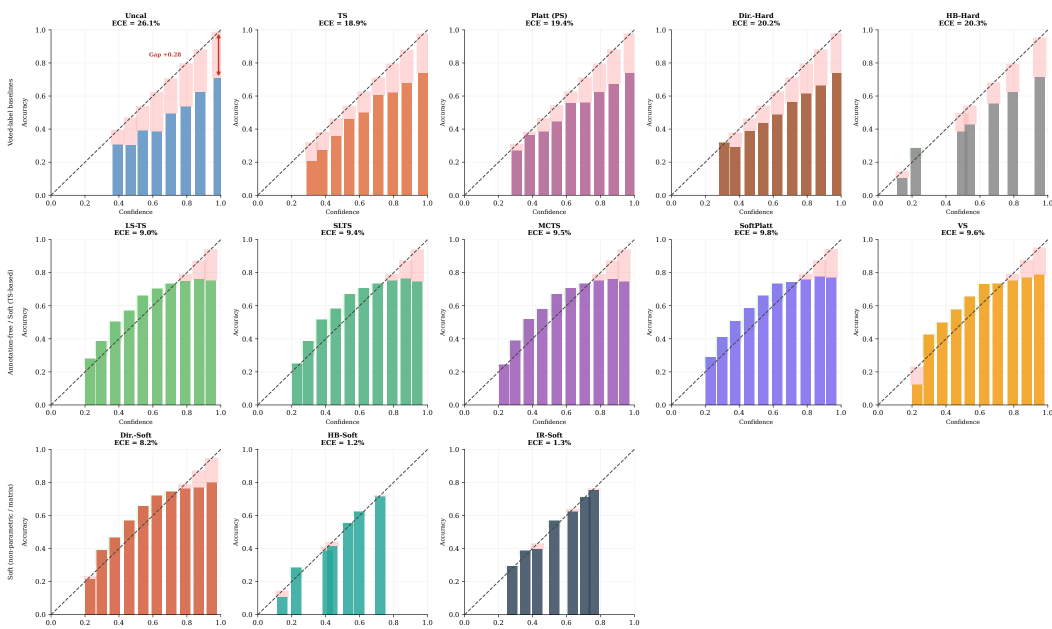

Figure 23: **Reliability diagrams (all methods) — ISIC 2019 EfficientNet-B4 (ECE$_{\text{true}}$).**

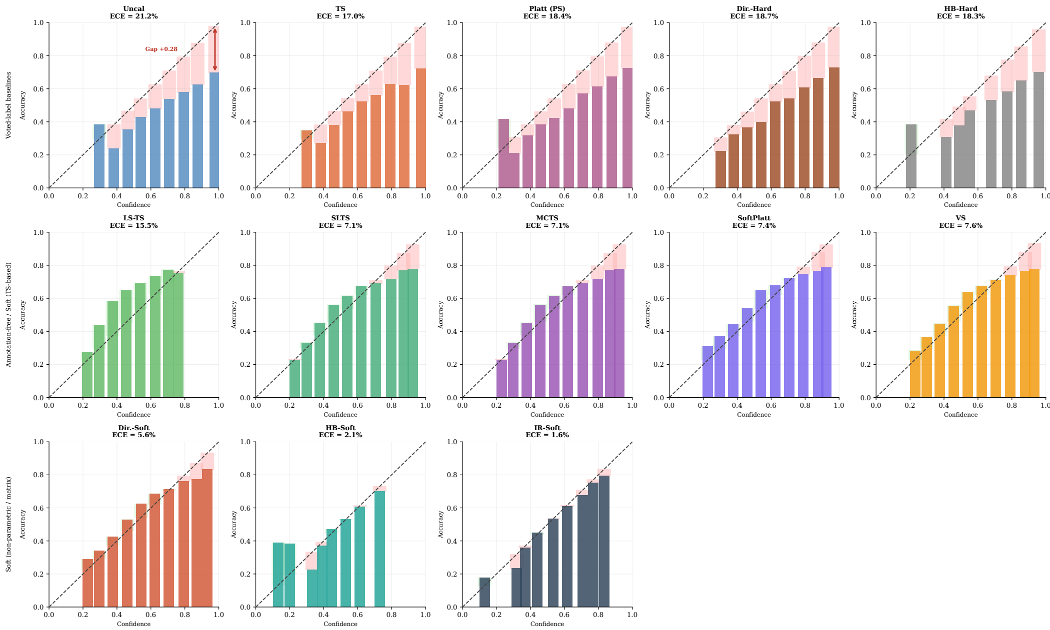

Figure 24: **Reliability diagrams (all methods) — ISIC 2019 ViT-S/16 ($\mathbf{ECE_{true}}$).**

DermaMNIST

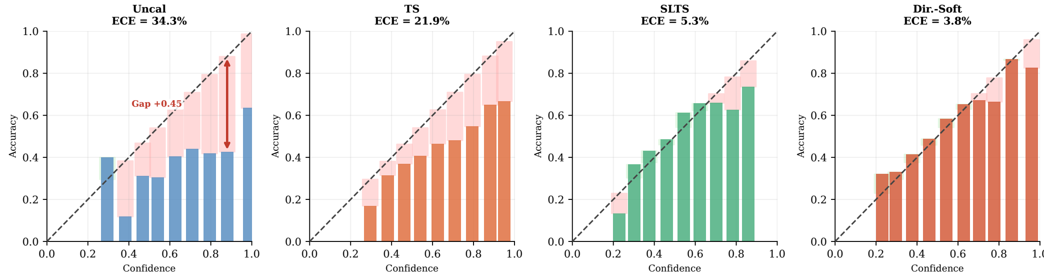

Figure 25: **Reliability diagrams (summary) — DermaMNIST ResNet-18.**

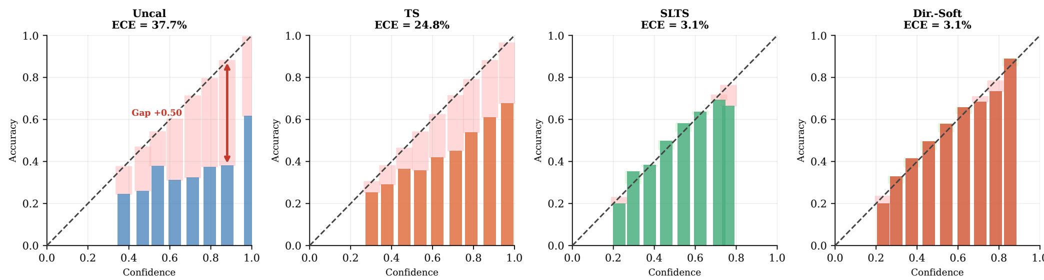

Figure 26: **Reliability diagrams (summary) — DermaMNIST ViT-S/16.**

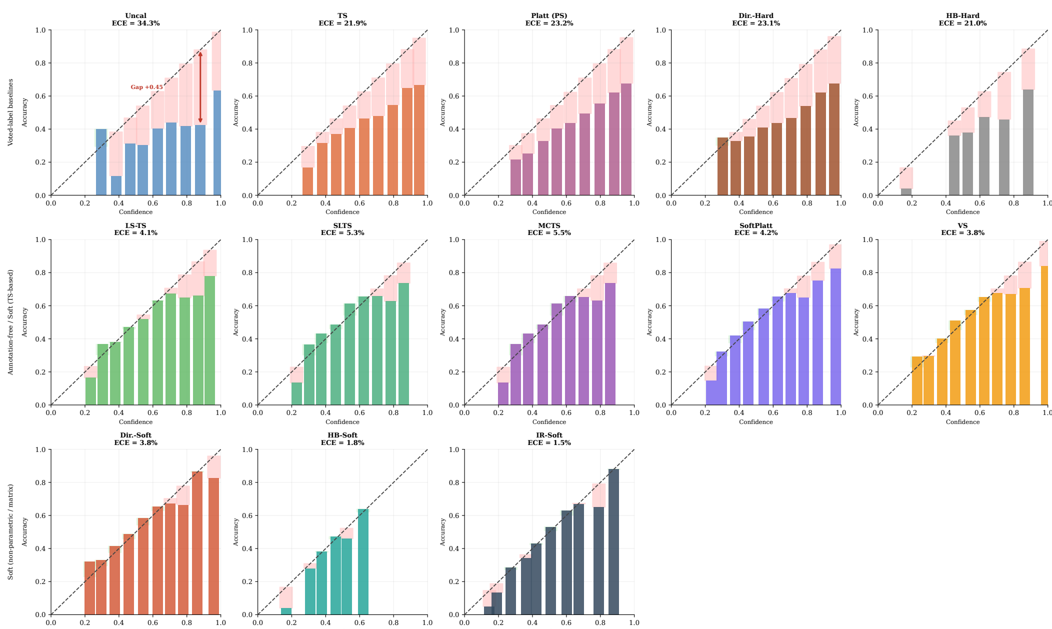

Figure 27: **Reliability diagrams (all methods) — DermaMNIST ResNet-18 ($\mathbf{ECE_{true}}$).**

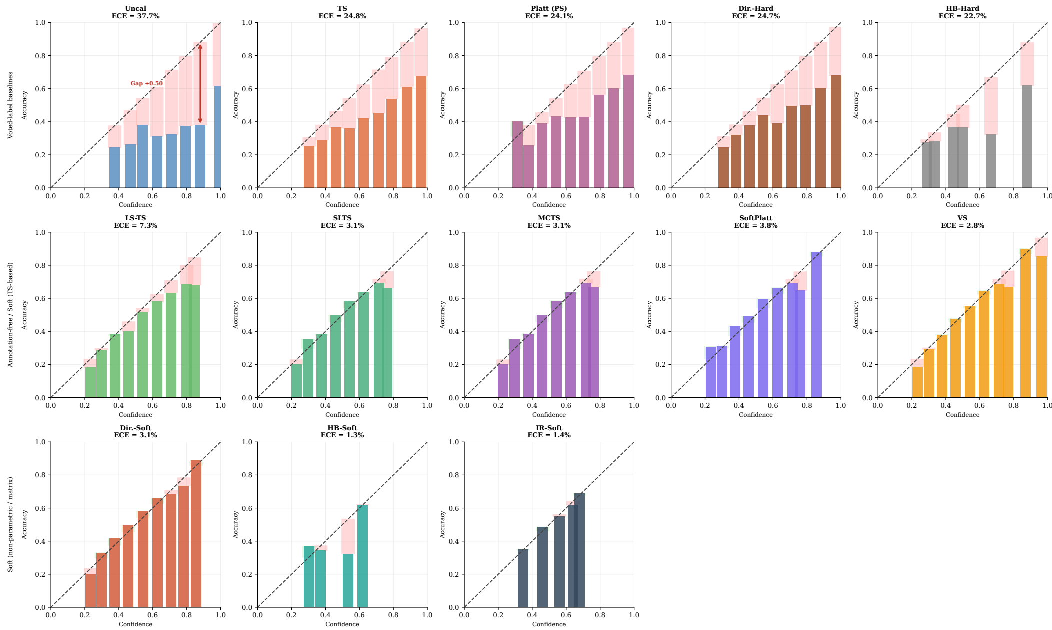

Figure 28: **Reliability diagrams (all methods) — DermaMNIST ViT-S/16 ($\mathbf{ECE_{true}}$).**

The diagrams confirm the quantitative results in Tables 1–2: TS reduces overconfidence relative to *Uncal* but leaves a large residual gap because it targets voted labels. Dirichlet-Hard performs similarly to TS and sometimes worse, confirming that the problem lies in the target, not model capacity. All soft methods substantially close the gap, with IR-Soft and Dir.-Soft achieving the most uniform residuals across all benchmarks.

I ISIC 2019 Annotator Confusion Matrix

Construction

Because the full per-image reader annotations of Liu et al. [2020] are not publicly available, we construct a synthetic annotator model using a clinically calibrated 8×8 confusion matrix C, where entry C_{ij} gives the probability that a board-certified dermatologist labels an image as class j when the consensus (majority-vote) diagnosis is class i. This confusion-matrix annotator model follows the framework of **?**, in which each annotator's behaviour is characterised by a class-conditional label-flipping distribution; it is widely used in crowdsourcing and multi-annotator learning [Rodrigues and Pereira, 2018, Uma et al., 2021].

Calibration sources. Diagonal entries (per-condition agreement rates) are set to match the per-condition inter-reader agreement reported in Liu et al. [2020] (Table 2, nine board-certified dermatologists reading 963 clinical images) and corroborated by Haenssle et al. [2018]. Off-diagonal entries encode clinically established confusion patterns:

- **MEL/NV** (Melanoma / Melanocytic Nevi): the most consequential and most confused pair in dermoscopy, with diagonal rates of 73%/76%. Off-diagonal entries $C_{\text{MEL,NV}} = 0.14$ and $C_{\text{NV,MEL}} = 0.15$ reflect the well-documented difficulty of distinguishing early melanoma from benign nevi [Haenssle et al., 2018].
- **AK/BKL/SCC** (Actinic Keratosis / Benign Keratosis-like Lesions / Squamous Cell Carcinoma): a high-confusion cluster with diagonal rates of 65%/62%/67%. The AK$\leftrightarrow$SCC confusion ($C_{\text{AK,SCC}} = 0.16$, $C_{\text{SCC,AK}} = 0.18$) is clinically significant because untreated AK can progress to SCC. BKL shares morphological features with both conditions.
- **DF/VL** (Dermatofibroma / Vascular Lesion): the easiest to distinguish, with diagonal rates of 87%/91%.

The mean diagonal agreement is $\bar{C}_{ii} = 75\%$, matching the aggregate inter-reader agreement reported by Liu et al. [2020] exactly.

Simulation procedure. For each test image x_i with consensus label y_i, we independently draw $m = 9$ synthetic annotations from the categorical distribution defined by row y_i of C:

$$a_i^{(k)} \sim \text{Categorical}(C_{y_i,\cdot}), \quad k = 1, \dots, m,$$

and set the empirical label distribution to $\pi(x_i) = \frac{1}{m} \sum_{k=1}^{m} \mathbf{e}_{a_i^{(k)}}$. This directly mirrors the $m = 9$ board-certified dermatologist annotation protocol of Liu et al. [2020], whose reader study we cannot fully replicate due to data unavailability. The resulting mean annotation entropy over the ISIC 2019 test set is $\bar{H} = 0.66$, reflecting the genuine diagnostic difficulty of the 8-class task.

Remark 2 (Class-conditional limitation)**.** *This synthetic annotator model is deliberately class-conditional: every image with the same consensus class y_i is assigned the same row $C_{y_i,\cdot}$ of the confusion matrix, regardless of its individual visual content. In practice, images within the same consensus class will differ in visual ambiguity, leading to instance-level variation in annotator disagreement that this model does not capture. Our experiments therefore evaluate calibration under a controlled form of clinically-informed ambiguity rather than true per-instance annotator distributions.*

Visualisation

Figure 29 shows the confusion matrix as a heatmap. The MEL/NV cluster (orange dashed box) and the AK/BKL/SCC high-confusion cluster (purple dotted highlights) are visually prominent. DF and VL have near-perfect diagonal entries and negligible off-diagonal mass.

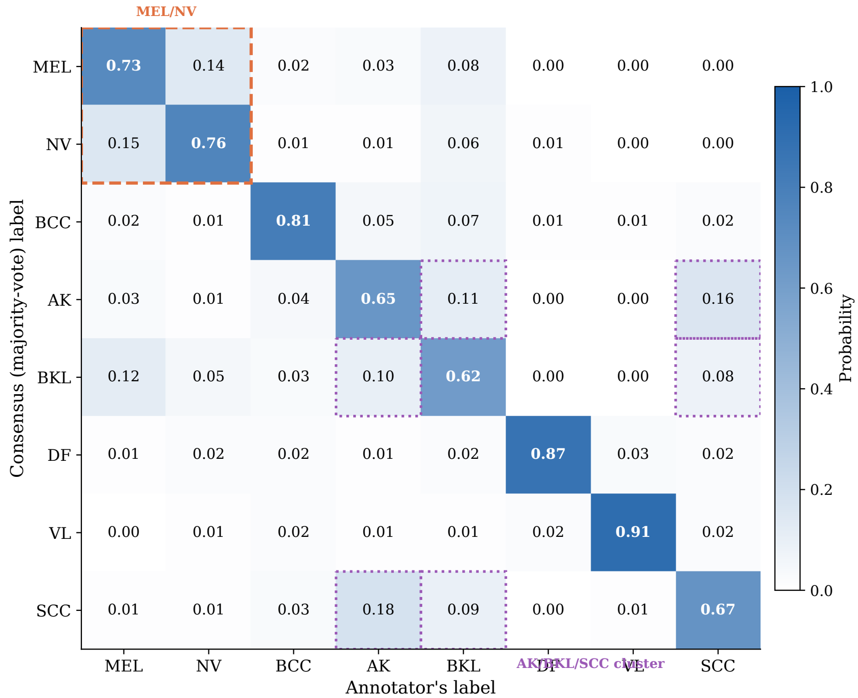

Figure 29: **Inter-reader confusion matrix for ISIC 2019.** Entry C_{ij} is the probability that a dermatologist labels an image as class j given the consensus label is i. Diagonal entries (bold) are the per-condition agreement rates, calibrated to match Liu et al. [2020] Table 2. Orange dashed box: MEL/NV most-confused pair. Purple dotted highlights: AK/BKL/SCC high-confusion cluster. Mean diagonal agreement $\bar{C}_{ii} = 75\%$.

Full confusion matrix

Table 5 lists the exact numerical entries of the confusion matrix.

Table 5: **Numerical values of the inter-reader confusion matrix for ISIC 2019.** Row = consensus (majority-vote) label; column = annotator's label. Each row sums to 1. Diagonal entries (bold) are per-condition agreement rates, calibrated to Liu et al. [2020] Table 2; mean diagonal agreement $\bar{C}_{ii} = 75\%$.

	MEL	NV	BCC	AK	BKL	DF	VL	SCC
MEL	**0.73**	0.14	0.02	0.03	0.08	0.00	0.00	0.00
NV	0.15	**0.76**	0.01	0.01	0.06	0.01	0.00	0.00
BCC	0.02	0.01	**0.81**	0.05	0.07	0.01	0.01	0.02
AK	0.03	0.01	0.04	**0.65**	0.11	0.00	0.00	0.16
BKL	0.12	0.05	0.03	0.10	**0.62**	0.00	0.00	0.08
DF	0.01	0.02	0.02	0.01	0.02	**0.87**	0.03	0.02
VL	0.00	0.01	0.02	0.01	0.01	0.02	**0.91**	0.02
SCC	0.01	0.01	0.03	0.18	0.09	0.00	0.01	**0.67**

J Empirical Validation of Proposition 2

Proposition 2 states that, under simplifying assumptions, the pointwise true-label calibration error of a voted-label-calibrated model is non-decreasing in the annotation entropy $H(x)$. Figure 30 provides empirical support: we bin test examples by their normalized annotation entropy $H(x)/\log K$ and plot the mean absolute error $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$ for TS and SLTS.

For CIFAR-10H (both architectures), TS error rises from $\approx 2\%$ in the near-unambiguous bin ($H/\log K \approx 0.02$) to $\approx 18\%$ in the high-ambiguity bin ($H/\log K \approx 0.48$), a clear monotone increase consistent with the proposition. SLTS also increases (the soft target itself becomes harder to match at high entropy) but does so more moderately. For ChaosNLI (DeBERTa-v3), which operates in a higher-entropy regime ($H/\log K \in [0.2, 0.9]$), TS again rises at the extremes while SLTS attenuates the error at moderate entropy levels. The crossover in the low-entropy bins reflects that SLTS over-smooths near-unambiguous examples (where voted-label and soft-label targets nearly coincide), consistent with the slight performance difference between TS and SLTS on unambiguous subsets.

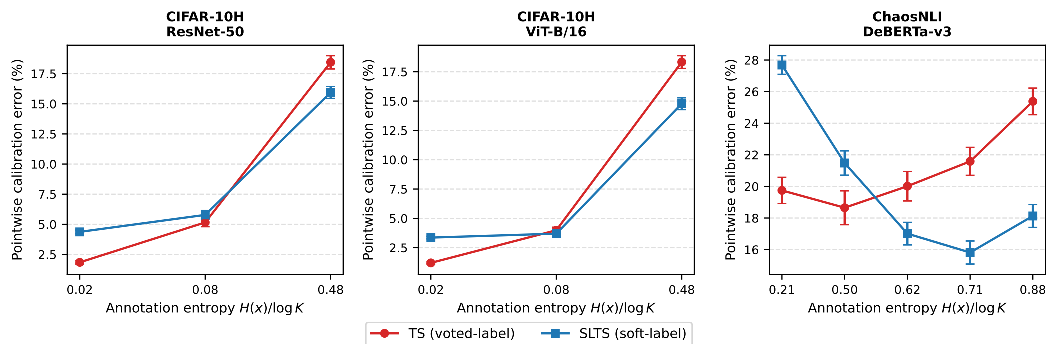

Figure 30: **Empirical validation of Proposition 2.** Test examples are grouped into equal-frequency bins by normalized annotation entropy $H(x)/\log K$. Each point shows the mean pointwise true-label calibration error $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$ (error bars: ± 1 s.e.). TS error consistently increases with annotation entropy on CIFAR-10H, confirming Proposition 2. SLTS moderates but does not eliminate this trend.

K LS-TS Ablation: Smoothing Strategy Comparison

We compare LS-TS against three simpler annotation-free smoothing baselines that share the same single-temperature family but differ in how the pseudo-target $\tilde{\pi}_i^{\mathrm{LS}}$ is constructed:

- **Fixed-LS** (ε=0.1): fixed data-independent smoothing weight, the standard label-smoothing value used in training [Szegedy et al., 2016].
- **Ent-LS**: per-instance $\varepsilon_i = H(\hat{p}(x_i))/\log K$, using normalised prediction entropy as a proxy for annotator disagreement.
- **CC-LS**: per-class $\varepsilon_k = \mathrm{mean}_{i:\, y_i^*=k}(1 - \hat{p}_{y^*}(x_i))$, accommodating class-level variation in model confidence (same idea as Vector Scaling applied to the smoothing weight).

All methods are evaluated on CIFAR-10H and ChaosNLI, the two benchmarks with real multi-annotator distributions, using the same experimental protocol as the main paper (seed 42, 50/50 stratified split).

Table 6: **LS-TS ablation: annotation-free smoothing strategies.** $\mathrm{ECE}_{\mathrm{true}}$ (%), Brier score, and NLL on CIFAR-10H and ChaosNLI. All methods use the same single-temperature family; only the pseudo-target construction differs. Fixed-LS uses ε=0.1 regardless of the data; Ent-LS uses per-instance entropy; CC-LS uses per-class mean complement-confidence. Best annotation-free method per column in **bold**.

	CIFAR R50			**CIFAR ViT**			**Chaos RoBERTa**			**Chaos DeBERTa**		
Method	ECE	Br	NLL	ECE	Br	NLL	ECE	Br	NLL	ECE	Br	NLL
TS (voted)	3.90	.1126	.346	4.01	.1043	.320	10.55	.537	.886	11.63	.549	.900
Fixed-LS (ε=0.1)	7.44	.1207	.332	6.00	.1058	.305	6.46	.525	.866	7.53	.536	.882
Ent-LS	3.16	.1116	.321	3.69	.1037	.308	3.17	.520	.862	8.58	.539	.885
CC-LS	**1.48**	**.1111**	**.296**	1.69	**.1016**	.279	4.15	.522	.873	**2.54**	**.529**	**.880**
LS-TS (ours)	**1.49**	**.1111**	**.296**	**1.62**	**.1016**	**.279**	**4.16**	**.522**	**.873**	2.65	.529	.881

Key findings. Fixed-LS with ε=0.1 consistently *under*performs or even *degrades* below TS (e.g., ECE 7.44% vs. 3.90% on CIFAR-10H ResNet-50), confirming that an arbitrary fixed smoothing weight is insufficient; the data-driven estimate of $\bar{\varepsilon}$ is critical. Ent-LS (per-instance entropy) improves over TS on most settings but is inconsistent: it works well on ChaosNLI RoBERTa-L (3.17%) but collapses on DeBERTa-v3 (8.58%), suggesting that prediction entropy is a noisy proxy for annotator ambiguity when models are already strongly confident. CC-LS (per-class ε) achieves results very close to LS-TS on all four settings (within 0.1 pp ECE), confirming that the global mean is a good summary of per-class behaviour for these benchmarks; the slight advantage of CC-LS on DeBERTa-v3 (2.54% vs. 2.65%) is within the single-seed variance reported in Appendix G.

L Adaptive Temperature Scaling with Voted Labels

A natural question is whether making the calibration map *adaptive* (predicting a per-instance temperature $T(x_i)$ rather than a global T^*) can close the gap between voted-label calibration and ambiguity-aware calibration, without requiring annotator data. We implement **ATS** [Joy et al., 2023]: a linear layer maps four logit-derived features $\phi(z_i) = [\max_k z_{ik},\ H(\mathrm{softmax}(z_i))/\log K,\ p_{(1)} - p_{(2)},\ p_{(1)}]$ to $\log T_i$ via $T_i = \mathrm{softplus}(w^\top \phi(z_i) + b) + 0.1$, trained with voted-label NLL identical to TS. The four features capture prediction scale, uncertainty, margin, and confidence. We apply ℓ_2 regularisation ($\lambda = 10^{-3}$) to avoid overfitting on small calibration sets.

Table 7: **ATS vs. TS and LS-TS across all 8 settings (annotation-free methods only).** $\mathrm{ECE}_{\mathrm{true}}$ (%) and Brier. ATS predicts a per-instance temperature from logit features but is trained with the same voted-label NLL as TS. **Bold** = best annotation-free method per column.

	C10H R50		C10H ViT		NLI RoBERTa		NLI DeBERTa		ISIC ENet		ISIC ViT		Derm R18		Derm ViT	
Method	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br
TS	3.90	.113	4.01	.104	10.55	.537	11.63	.549	18.54	.560	16.59	.617	23.05	.686	24.36	.683
ATS	4.10	.113	4.10	.105	11.36	.538	13.09	.552	18.83	.560	17.53	.631	22.82	.687	25.35	.686
LS-TS	**1.49**	**.111**	**1.62**	**.102**	**4.16**	**.522**	**2.65**	**.529**	**9.34**	**.528**	**15.49**	**.619**	**5.05**	**.630**	**7.36**	**.615**

Finding. ATS fails to improve over TS in all 8 settings and in several cases degrades it (ECE rises from 11.63% to 13.09% on DeBERTa-v3; from 16.59% to 17.53% on ISIC ViT-S/16). The per-instance temperature learned from logit features adapts to *model uncertainty*, not to *annotator disagreement*; these two quantities are loosely correlated at best (cf. Figure 30). In contrast, LS-TS, which uses the same global-temperature architecture as TS but simply shifts the calibration target toward a smoothed distribution, reduces ECE by 7–78% across all 8 settings without any annotator data. This confirms that the calibration target is the critical design axis: added architectural flexibility without the right supervision signal yields no benefit.

M Implementation Details

CIFAR-10H models. *ResNet-50*: ImageNet-1k weights (`ResNet50_Weights.IMAGENET1K_V1`); FC layer replaced with 2048×10 linear; AdamW, lr$= 10^{-4}$, weight decay 10^{-4}, batch 128, cosine annealing, 30 epochs, 224×224. *ViT-B/16*: ImageNet-21k weights; classification head replaced; AdamW, lr$= 5 \times 10^{-5}$, weight decay 10^{-4}, batch 64, cosine annealing, 20 epochs, 224×224.

ChaosNLI models. *RoBERTa-Large*: `roberta-large-mnli` from HuggingFace, pre-fine-tuned on MultiNLI [Liu et al., 2019]; no additional fine-tuning. *DeBERTa-v3-base*: `cross-encoder/nli-deberta-v3-base` from HuggingFace [He et al., 2023]; no additional fine-tuning. Both models use the combined ChaosNLI SNLI+MNLI subset ($n = 3{,}113$), split 50/50 into calibration and test (stratified, seed 42).

ISIC 2019 models. *EfficientNet-B4*: ImageNet-1k weights; stratified 70/15/15 split; class-weighted cross-entropy (NV: $\approx$67% of training images); AdamW, lr$= 3 \times 10^{-4}$, weight decay 10^{-4}, 2-epoch linear warm-up + cosine decay, 20 epochs, 380×380. *ViT-S/16*: ImageNet-21k weights; same split and loss weighting; AdamW, lr$= 10^{-4}$, weight decay 10^{-4}, 2-epoch warm-up + cosine decay, 20 epochs, 224×224.

DermaMNIST models. *ResNet-18*: ImageNet-1k weights, class-weighted cross-entropy, AdamW, lr$= 10^{-4}$, 30 epochs. *ViT-S/16*: ImageNet-21k weights, same training protocol, 224×224.

Calibration optimizers. TS, SLTS: LBFGS, lr$= 0.1$, 500 iterations, tolerance 10^{-9} (gradient), 10^{-11} (loss). Platt/VS: Adam, lr$= 0.01$/0.05, weight decay 10^{-4}, 2000 steps.

ECE bins. $B = 15$ equal-width bins in $[0, 1]$; bins with fewer than 1 example are excluded.

Compute. CIFAR-10H fine-tuning: $\approx$20 min on one NVIDIA A100. All calibration methods: < 60 s on CPU after logit caching.